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Product Description

PGW, SMF, and UPF

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1. General Information

floLIVE's Connectivity Management Platform (CMP) is a cloud-based solution that encompasses a fully virtualized 3G/4G/LTE and 5G core network as well as a complementary suite of Business Support Systems (BSS) that enable our customers to take full ownership over their IoT business, be it device manufacturing and distribution or complete IoT connectivity management.

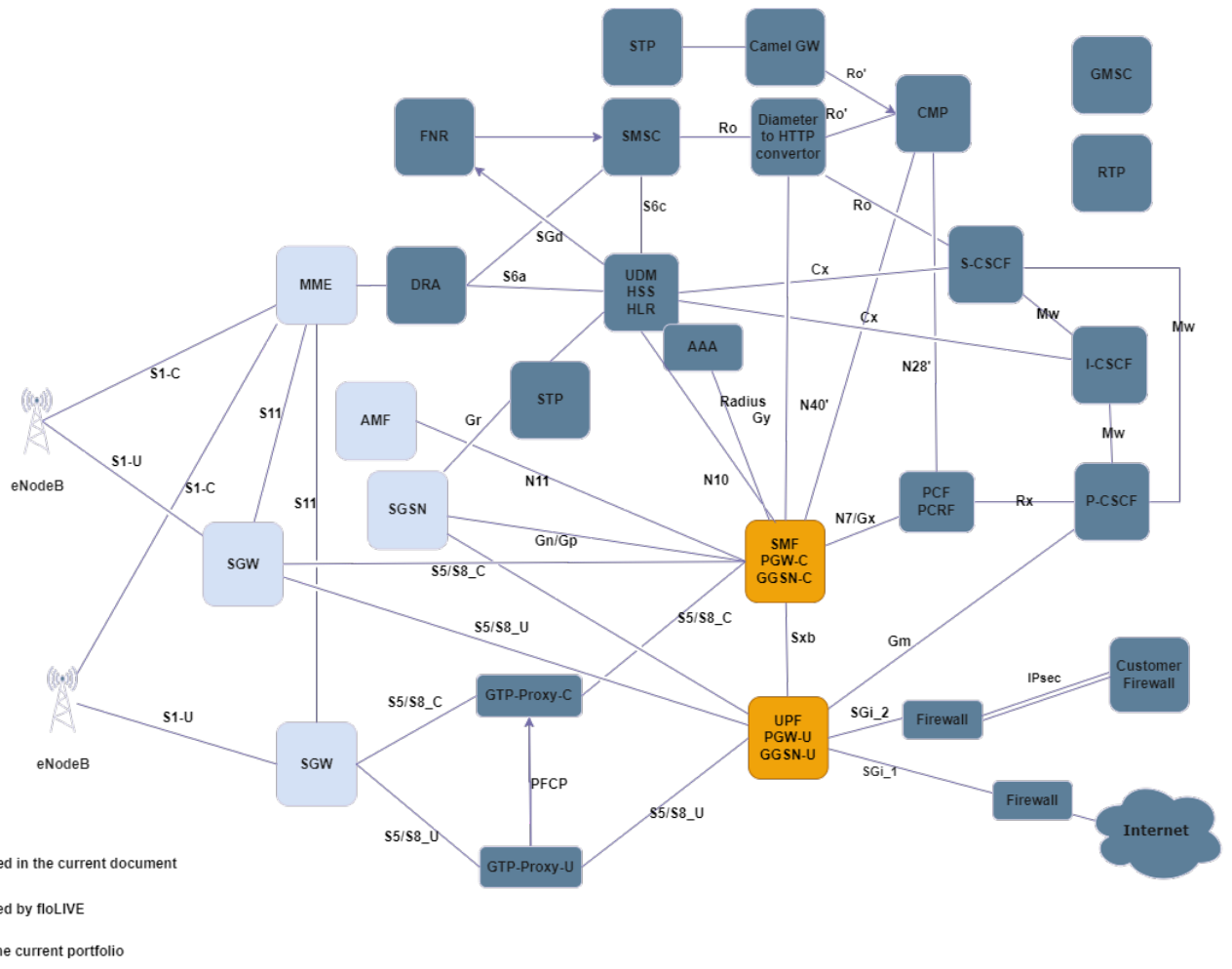
The platform provides the functionality necessary to support various workflows in managing the connectivity aspects of modern IoT and M2M device arrays.

floLIVE's CMP-as-a-Service solution comprises several subsystems and management applications.

The subsystems in use are the following:

- **floCONNECT Connectivity Management** - used for setting and managing accounts, users, SIMs, Self-care, Rules & Alerts, and more.
- **HLR/HSS** - Used for managing subscriber profiles, RAT Types, IMSI/MSISDN ranges and more.
- **GGSN/PGW (UPF & SMF)** - used for configuration and management of APNs, IP Pools, routing tables, etc.
- **SMSC** - used for configuration of SMS logic, forwarding rules and policies, external SMPP connections, and more.
- **floSIM Multi IMSI SIM Management** - used for configuration, management and control of SIM cards, business logic, PLMN definitions, OTA transactions, key management, and more.

Figure 1 - General Information



2. Overview

FloLive's SMF/UPF product supports 2G,3G,4G, 5G-NSA, and 5G SA. On top of SMF/UPF for 5G-SA, it also combines PGW and GGSN functionality into a single product. Unless otherwise specified, SMF denotes the control-plane functions PGW-C and GGSN-C, while UPF denotes the user-plane functions PGW-U and GGSN-U.

This document outlines the architecture and connectivity framework for the User Plane Function (UPF) and Session Management Function (SMF) within a mobile network. With increasing demands for data accuracy, the PFCP protocol plays a pivotal role in ensuring that precise information is reliably transmitted to the designated Data Network from the outset.

The SMF is integral to defining a scheduled control mechanism, which the UPF subsequently enforces on the N4 interface, aligning with the user-defined policies. The PFCP protocol facilitates seamless data transmission across various interfaces, enhancing the UPF's capabilities in session-based connection management and enabling systematic traffic routing between N3 and N6 interfaces.

3. Architecture

Flo Live uses CUPS architecture, separating the flow into control and user plane, respectively, to UPF and SMF Nodes in the 5G solution.

3.1. Classic Architecture

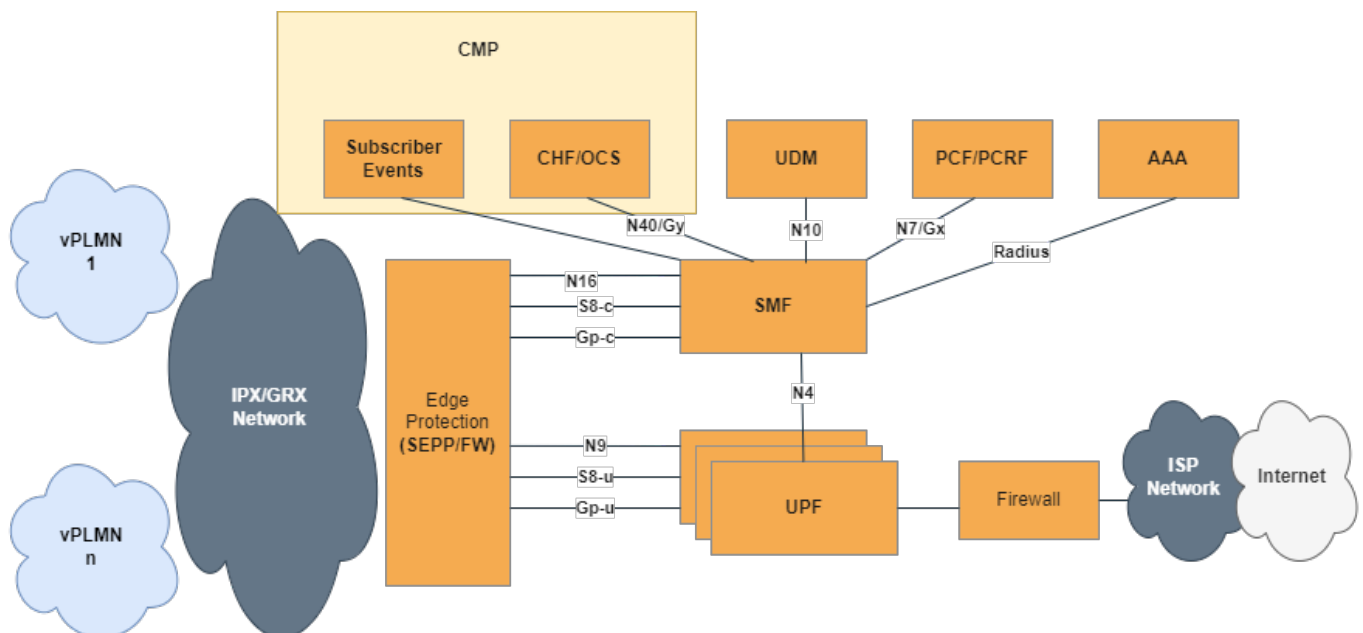
This architecture connects multiple vPLMNs to FloLive's edge via IPX/GRX networks. At the edge, robust security is enforced using a firewall for both GTP control and user plane traffic, alongside a SEPP (Security Edge Protection Proxy) that safeguards SBI interfaces such as N16/Nsmf.

The SMF is integrated with various core elements through standardized interfaces: it communicates with the CHF/OCS over N40 and Gy interfaces respectively, connects to the UDM via the N10 interface, interfaces with PCF/PCRF via N7 and Gx, and links to AAA servers using the RADIUS protocol. In addition, the SMF manages multiple UPFs through the N4 interface.

Each UPF is connected to external networks (including the internet and customer sites) through a firewall that supports NAT functionality and secure VPN tunnels. FloLive's architecture follows 3GPP 5G Standalone (SA) and Control and User Plane Separation (CUPS) principles. This ensures a clear demarcation between control and user planes — both in the 5G system (via SMF and UPF) and in legacy nodes (via control and user-plane separation in PGW and GGSN).

The diagram below illustrates the classic architecture employed by FloLive for its GGSN, PGW, SMF, and UPF components.

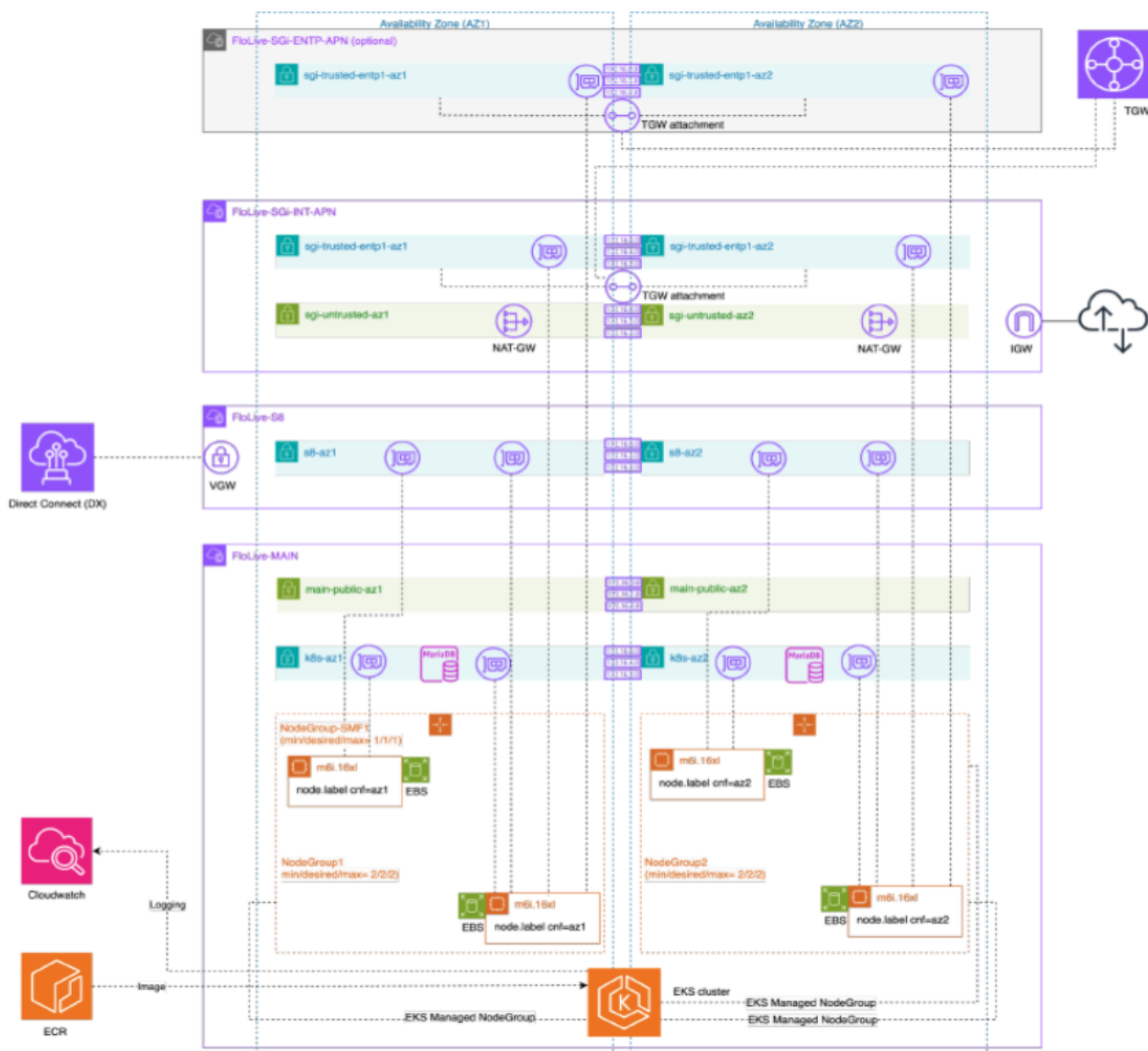
Figure 1 - Classic Architecture



3.2. Classic Architecture on Public Cloud Infrastructure

The floLIVE SMF/UPF solution is fully cloud-native and can be seamlessly deployed on public cloud infrastructures, leveraging a multi-VPC architecture for performance, scalability, and security. In this setup, one VPC is dedicated to handling S8 interfaces, facilitating interconnectivity with legacy 4G cores or roaming partners. This VPC is directly connected to the IPX/GRX provider via AWS Direct Connect, ensuring low-latency, secure, and high-throughput communication. A separate VPC is allocated for internet breakout, equipped with NAT Gateways and Internet Gateways (IGWs) to securely route user-plane traffic to the public internet. Another VPC serves enterprise connectivity, utilizing IPsec tunnels or AWS Direct Connect fabric links to establish secure communication with customer on-premises networks. Finally, a central VPN VPC is deployed to manage internal control-plane and service-to-service traffic, maintaining logical isolation and enforcing security policies. This architecture provides a highly modular, secure, and resilient deployment model aligned with the dynamic needs of modern mobile core networks.

Figure 2 - Classic Architecture on Public Cloud



3.3. Classic SMF/UPF on Prem Infrastructure

A Kubernetes cluster with 3 nodes is needed to install the SMF/UPF locally.

The node's resources are subject to dimensioning per the traffic model number of active subscribers.

As a reference, the following setup is recommended:

Each site has its own Kubernetes cluster, with three hosts.

All hosts must agree to the following conditions:

- 32 vCPU cores per node
- 128 GB memory (min)

- 2 X 240 GB SSD in RAID 1
- 2 NICs, 10 GB

3.4. N4/Sxb Communication between SMF and UPF

1. SMF utilizes the Packet Forwarding Control Protocol (PFCP) to provide rules to the UPF.
2. UPF establishes a GTP-U channel based on these rules.
3. Forwarding data using IPv6 and IPv4 PDU Session types.
4. The UPF forwards the data, but N4/SXb is only involved in the first point.

An endpoint on a device cannot always stay operational. UPF can review and troubleshoot all historical connectivity and service issues in a report.

You can find the report types in the table below.

Attribute	Description
Report type	• Downlink Data Report • Usage Report • Error Indication Report • User Plane Inactivity Report
Downlink Data Report	Report that UPF has received and buffered a downlink packet
Usage Report	Several Usage Reports can be included
Error Indication Report	Report that an error indication is received from a remote GTP-U endpoint
Load Control Information	Load metric providing SMF with information about UPF load level
Overload control Information	UPF includes the information during an overload condition
Additional usage report information	Total number of usage reports that are sent in PFCP Session Report Request

3.5. Resource and Capacity Requirements

Review the resource and capacity requirements for PGW-C/SMF in the table below.

K8s POD	Resources for Production Deployment	Capacity
PGW-C / SMF	CPU: 5 Memory: 15 GB	
SMF API	CPU: 0.3 Memory: 0.75 GB	

K8s POD	Resources for Production Deployment	Capacity
SMF UI	CPU: 0.3 Memory: 0.5 GB	
Total per SMF	CPU: 17 Memory: 174 GB Hugepages: 228 MB	The SMF capacity depends on the specific call model and is constrained by the following factors: 1. Up to 1 000 000 simultaneous sessions 2. Up to 2 000 TPS on S5/S8 interface 3. Up to 1 000 000 active users in busy hour

Review the resource and capacity requirements for PGW-U/UPF in the table below.

Node	Resources for Production Deployment	Capacity
PGW-U / UPF	CPU: 5 Memory: 15 GB	
UPF API	CPU: 0.3 Memory: 0.75 GB	
UPF UI	CPU: 0.3 Memory: 0.75 GB	
UPF Config	CPU: 0.3 Memory: 0.5 GB	
PGW-U/UPF	CPU: 5 Memory: 4 GB Hugepages: 228 MB	The UPF capacity depends on the specific call model and is constrained by the following factors: 1. Up to 1 000 000 simultaneous sessions 2. Up to 2 000 TPS on S5/S8 interface 3. Up to 1 000 000 active users in busy hour 4. Up to 2.5 Gigabits per second (Gbps) aggregate throughput (uplink+-downlink)

3.6. Available Integrations

3.6.1. SGSN

The SGSN integration can be done directly to the PGW or via the GRX network, in case the integration is internal there is no IR21 IP required, the MNO needs to define the relevant APN on it's DNS and assign the PGW IP. This IP needs to be reachable from the MNO network and the proper routing needs to be assigned on both sides.

In case of routing via the GRX, the IP needs to be a known IP from IR21, these IPs will be published on the local BGP network connected to the GRX, so all traffic to these IPs will be routed over the GRX to the local PGW.

3.6.2. AAA / Radius

AAA for Static IP can support two scenarios:

1. The local AAA on the PGW site can serve all the customers using the PGW.
2. Integration with customer AAA. The IP range that can be used will be assigned by the IP manager of the local IP site and should be followed by the customer. The AAA server needs to be reachable, and usually, an IPsec VPN to the customer is required.

3.6.3. GRX

The local PGW must connect to each GRX for the required customer. In case there is a request to support several GRX connections, each one of GRX will have a dedicated PGW service in the Kubernetes cluster.

3.6.4. ISP

At least one ISP with BW of the traffic estimated needs to be connected to the PGW, it is better to have 2 for redundancy.

3.6.5. IR21

In case the PGW needs to be reached for the GRX, then real IPs that are published on the IR21 must be assigned to OCR/CDR.

3.6.6. OCS / CDR

Each PGW has local CDR files. These are offline CDRs that can be used for reconciliation and transferred to customers (need to filter the relevant CDRs from the files). The CDRs are not rated.

Gy interface (Radius/Diameter) is supported and can be routed to the OLC system for MVNO and customers.

3.6.7. PCRF / PCF

PCRF can be installed locally and integrated to the PGW.

3.6.8. DNS

There are two types of DNS required for the solution:

1. For APN resolution. Usually, this DNS is handled by the MNO but in the case of wildcard APNs, the MNO can route DNS requests to a local DNS.
2. DNS for the SGI traffic. This should be a public DNS and needs to be defined per APN in the PGW.

3.7. Data Connectivity

3.7.1. Mobility and Data Connectivity

Data is transported between the radio access network and the UPF over the **N3** reference point.

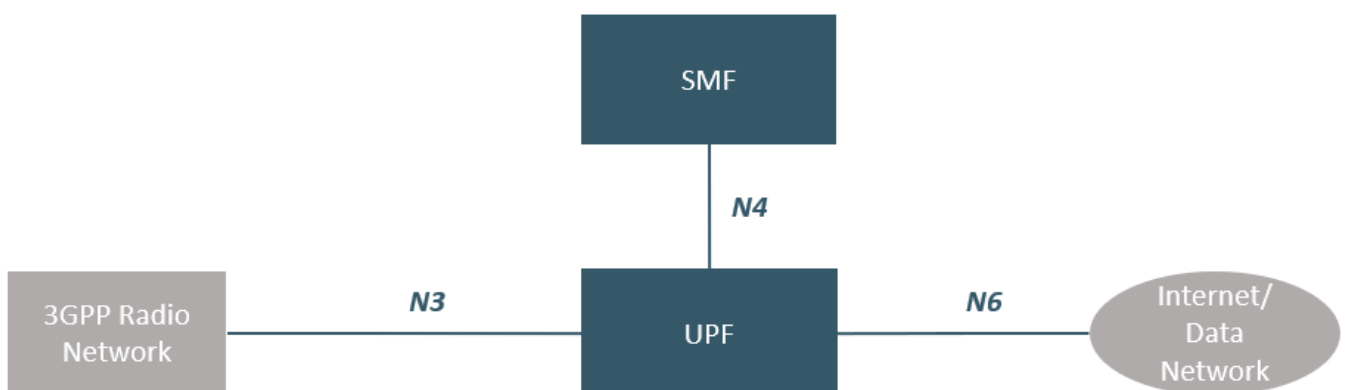
IP routing is done on the channel header IP address instead of the equipment IP address.

The **N3** reference points expand on the new ways of managing the Quality-of-Service of specific data in the Evolved Packet Core (EPC) Architecture design by mapping data flows to channels.

Possibilities to use **N6** virtual private networking create secure channels and data transfers.

IP packets are routed based on the actual IP address of the device - the traffic is not tunneled.

Figure 3 - UPF Connection to External Data Networks



Managing a device signaling is done through SMF/UPF dual traffic relation - UPF control is per requested data session, in which SMF can specify to create or limit information:

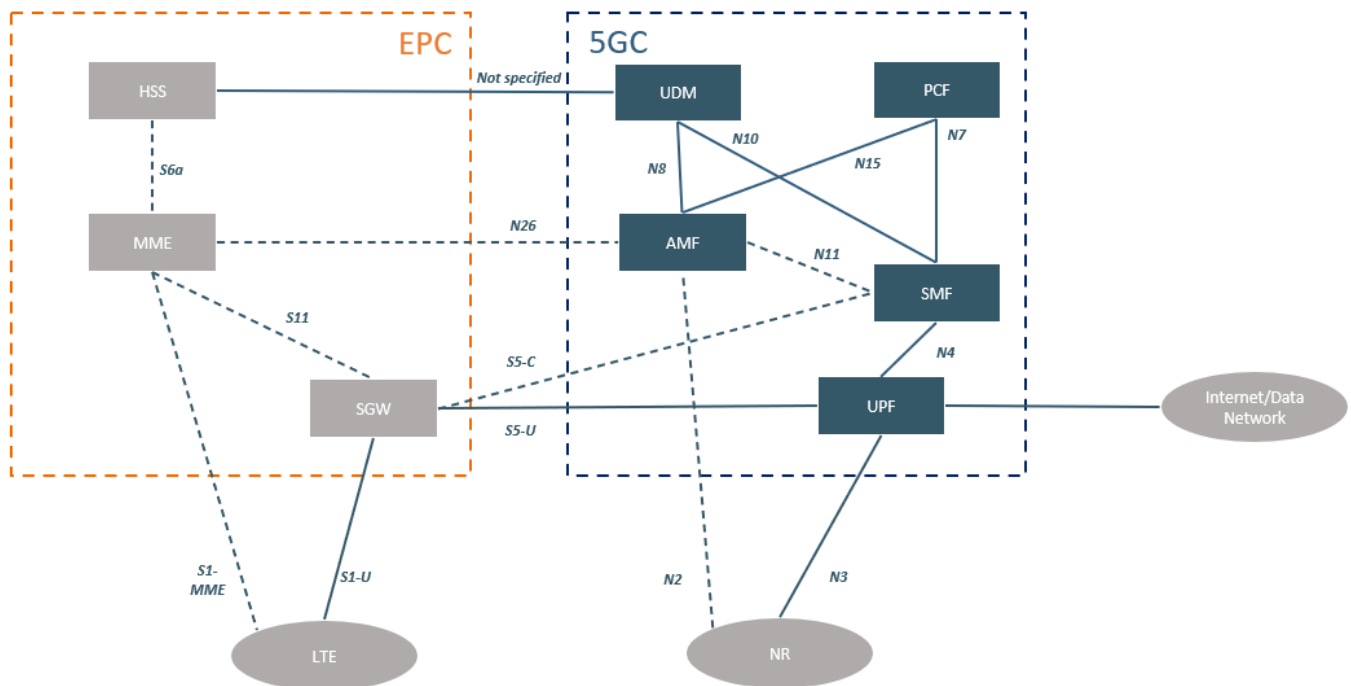
- Traffic detection rules

- Packet forwarding rules
- Providing Quality-of-Service parameter values for data flows

3.7.2. Compatibility

The forwarding usage based on SMF logic and User Plane Functionality (UPF) should be specified by using the Control and User Plane Separation (CUPS) strategy across the S5-C and S5-U interfaces. The UPF ensures full compatibility and control between New Radio and LTE PGW.

Figure 4 - Compatibility



Note: Existing 4G networks do not need to be migrated to the 5G architecture.

3.7.3. Roaming

To make a device function properly, even though it is in out-of-home network coverage area **N9** adjustment is allowed to be set. The UPF can reconnect to IP targets and route them back to the needed point.

This recognition can be determined in the following order:

- Once a device is registered in the Visited Network, immediate traffic forwarding to the Home Network is done.
- A reactive check and inter routing data key roaming unlock track is pinpointed to the existing UPF.
- A device can operate by using resources from two public mobile networks - a Visited Network where the device is located and a Home Network.

The **N9** is a roaming reference point that allows:

- To be used for carrying the requested data between the networks.
- To connect to the Internet or other IP services that are in the home network, in which the device is in the visited network.

Figure 5 - Local Breakout of the Data

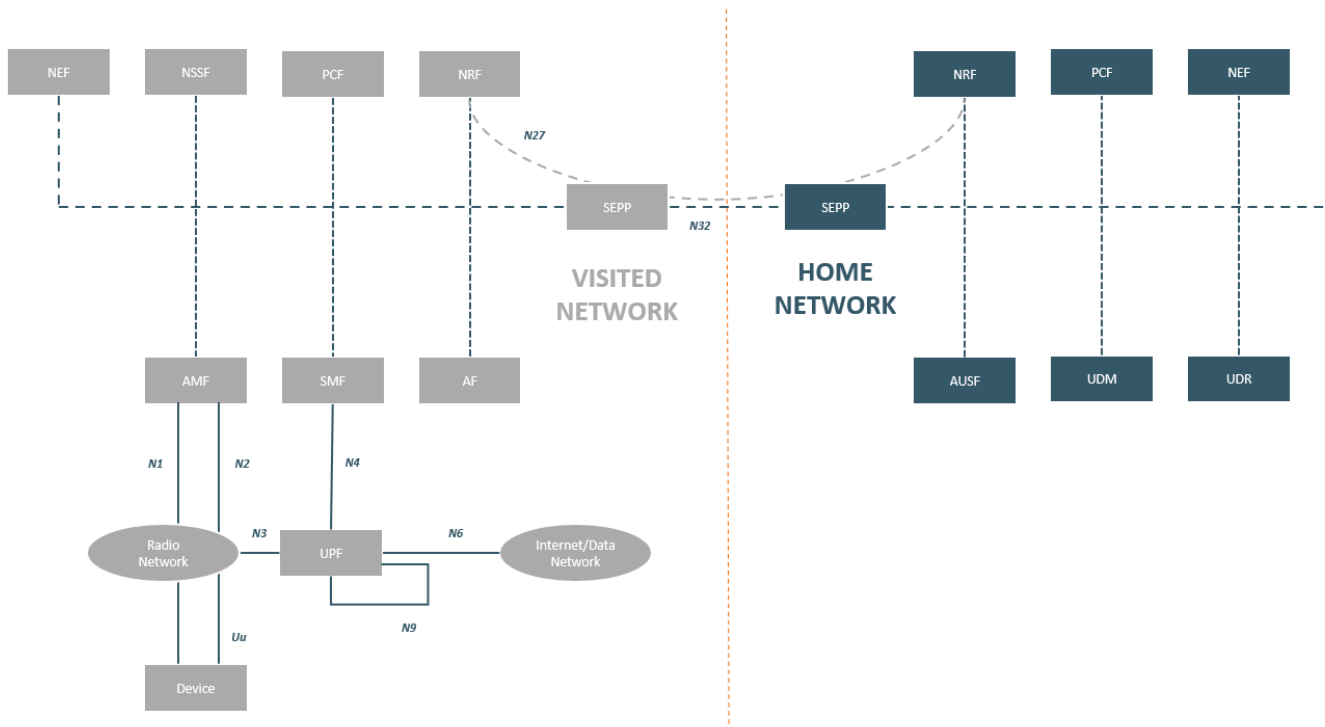
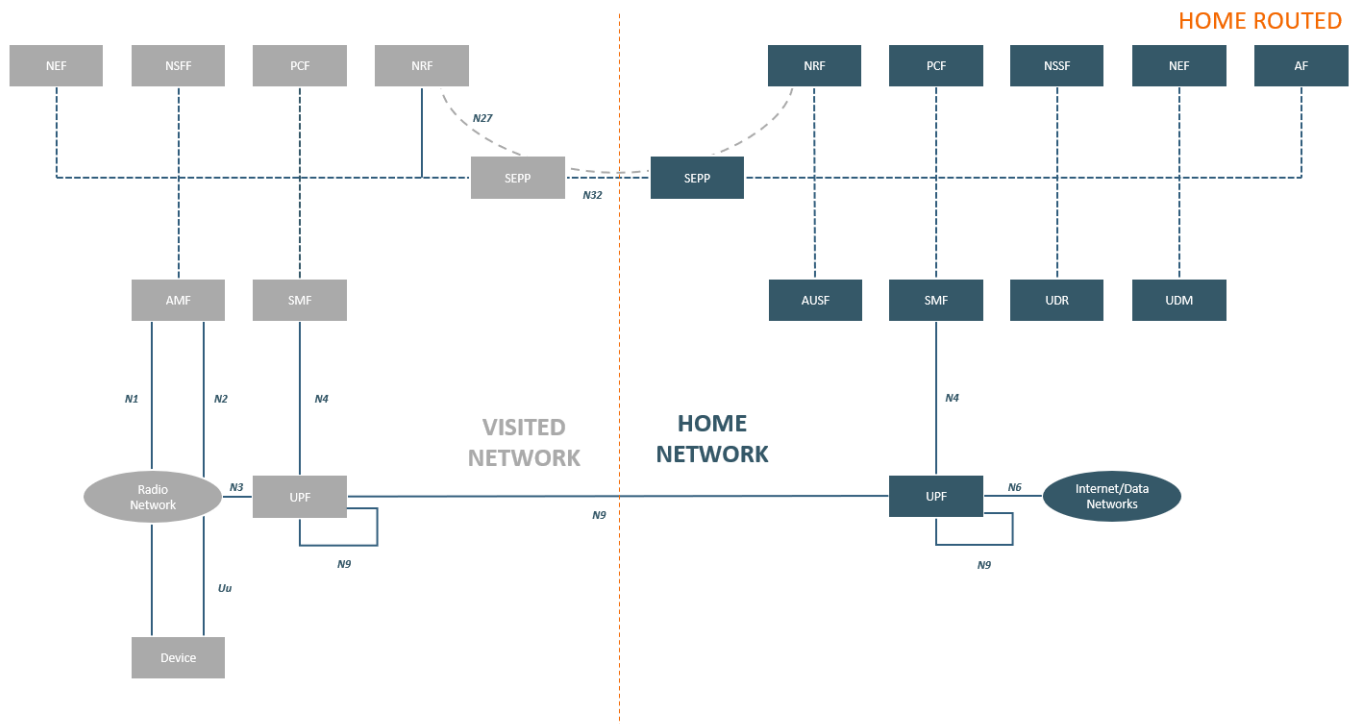


Figure 6 - Routing of Data to Home Network



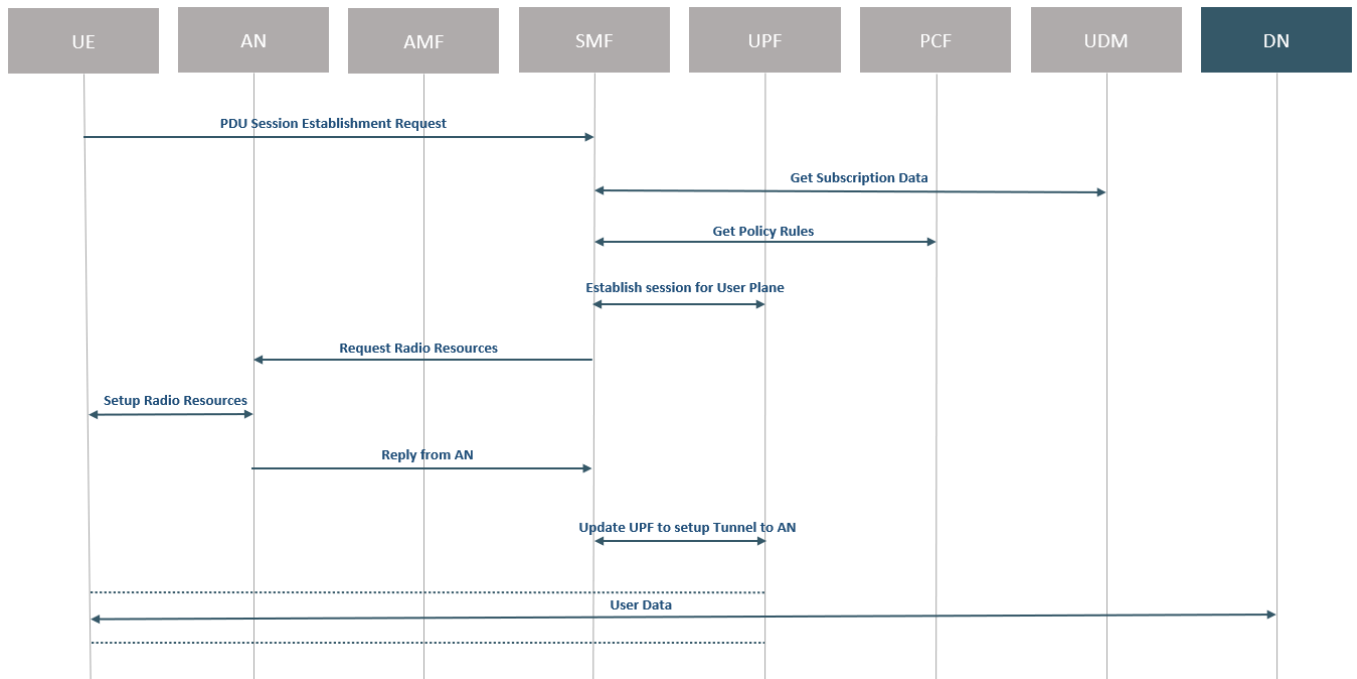
UPF is making interconnections to Visited and Home Networks utilizing the N9 reference point that can be set between out-of-reach global servers.

3.7.4. Connectivity Service to Data Network

Establishing a Data Network (DN) involves creating a user plane connection between a device and an active DN. This is done by requesting a PDU session through the UPF to establish an end-to-end point with a predefined name.

1. To allow access to network resources, it is facilitated through the application layer, which in the context of 5G is determined by the Protocol Data Unit (PDU) layer.

Figure 7 - Session Establishment Procedure

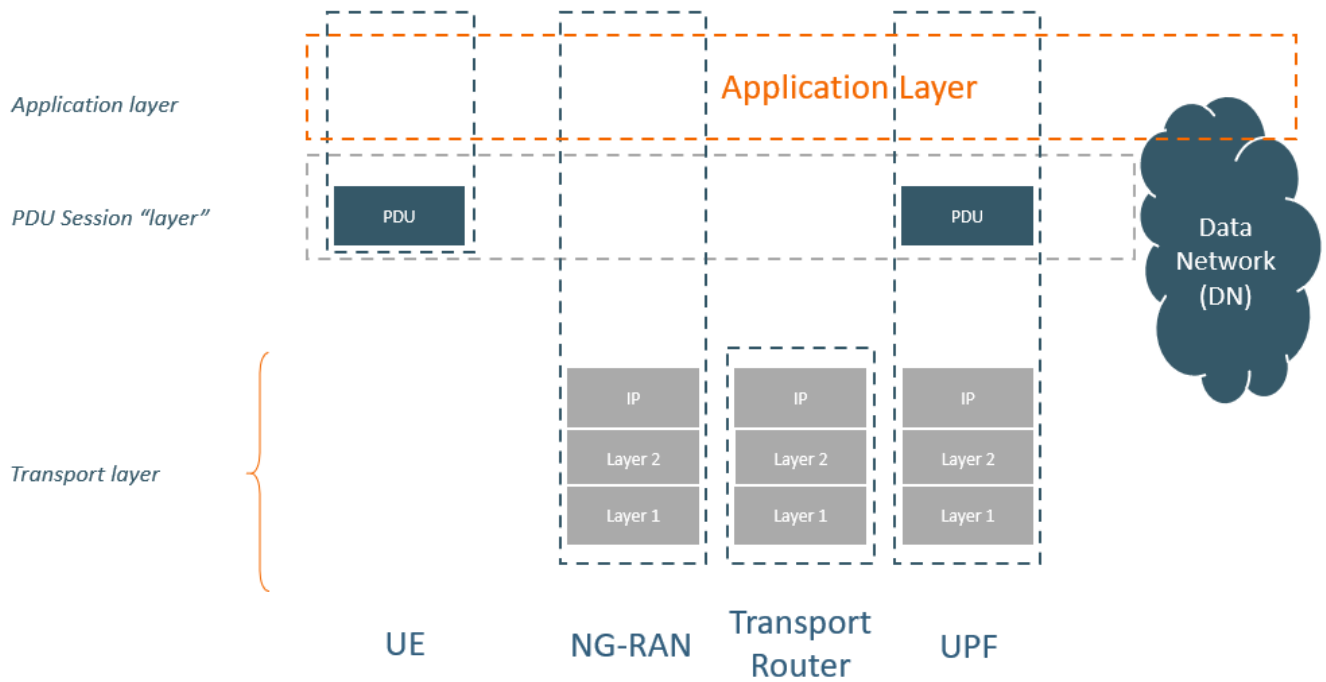


2. The User Plane connection (the PDU Session) is separated from the transport connection between network nodes, in which UPF supports data forwarding for:

- Centralization is set for the upper protocol layer
- Per-Point Security
- Mobility
- Accounting
- Quality of Service

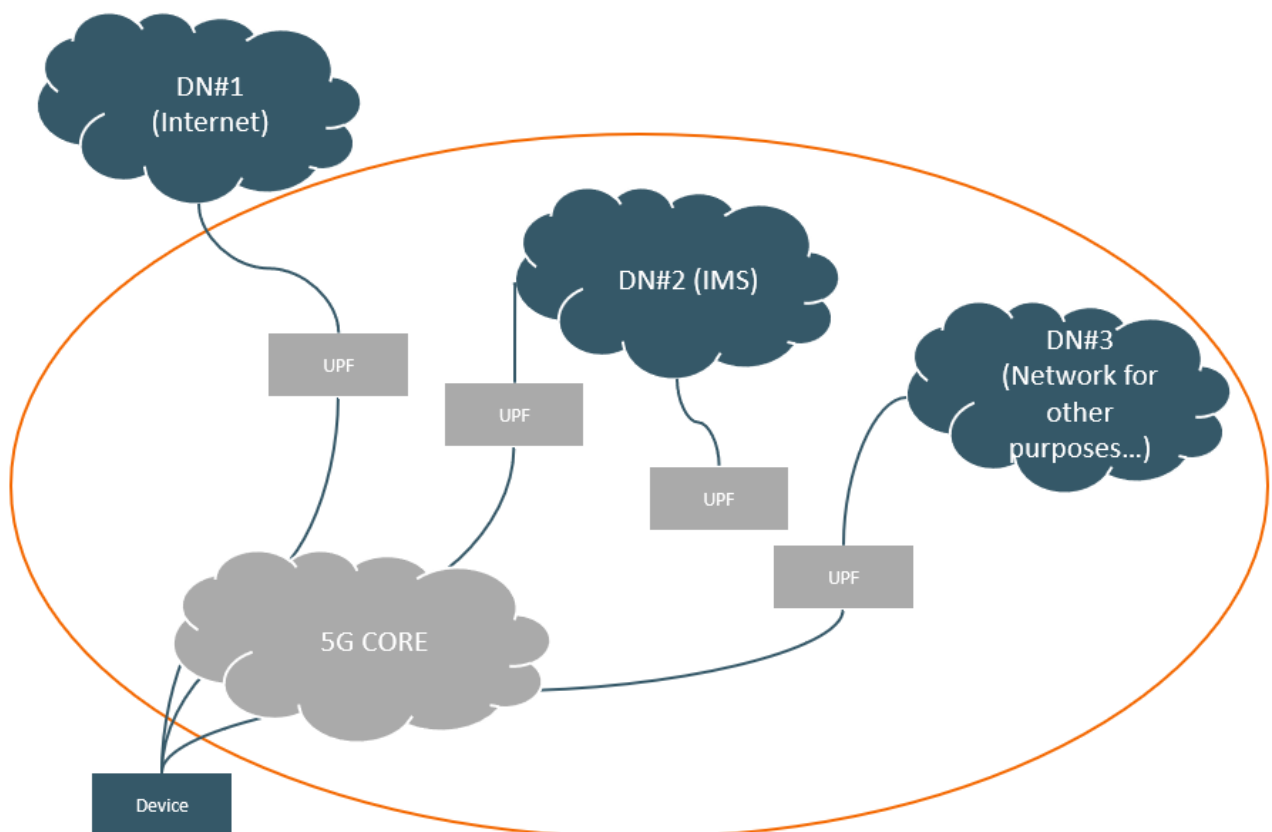
NOTE: All rounded flexible functionality allows deployment of any transport technology independently of the Application layer.

Figure 8 - User Plane Protocol Stack



Very useful connectivity comes where a multiple PDU Session is required to be set in parallel in case a device requires simultaneous access to the Internet and IP Multimedia Subsystems IMS services.

Figure 9 - UPF Connect to Multiple DNs



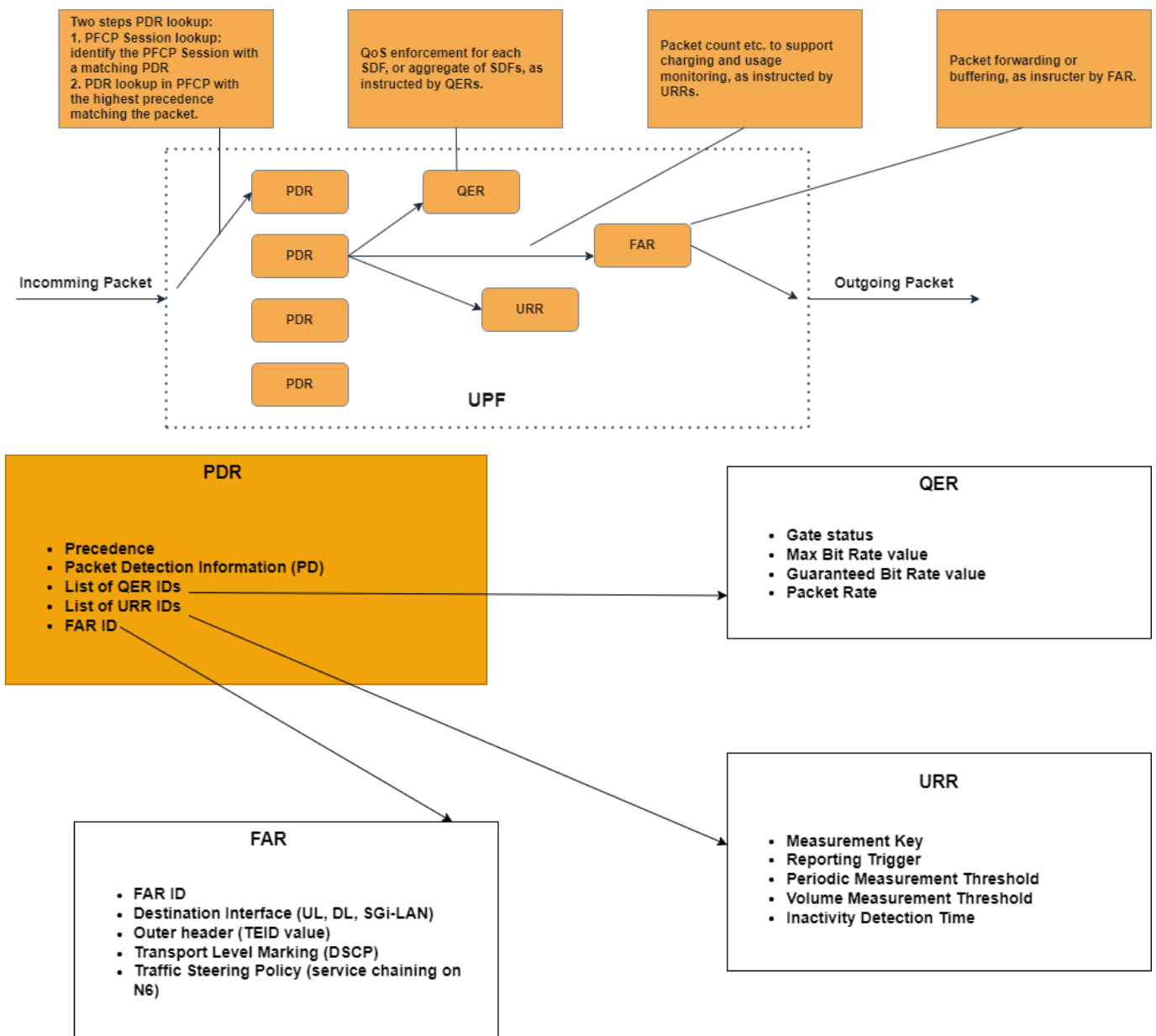
NOTE: Available types: IP-based PDU Session types: IPv4, IPv6, and dual-stack.

3.7.5. Packet Forwarding Model and PFCP Rules

Dedicated control on PFCP session is making UPF processing by utilizing:

1. **Packet Detection Rule (PDR)** detects and classifies incoming user data traffic (PSUs) - IP filter.
2. **Quality-of-Service Enforcement Rule (QER)** enforces QoS-bit rate parameters.
3. **Usage Reporting Rule (URR)** measures packets and bytes and reports to SMF.
4. **Forwarding Action Rule (FAR)** forward towards DN.

Figure 10 - Processing Model and PFCP Rule and Relations

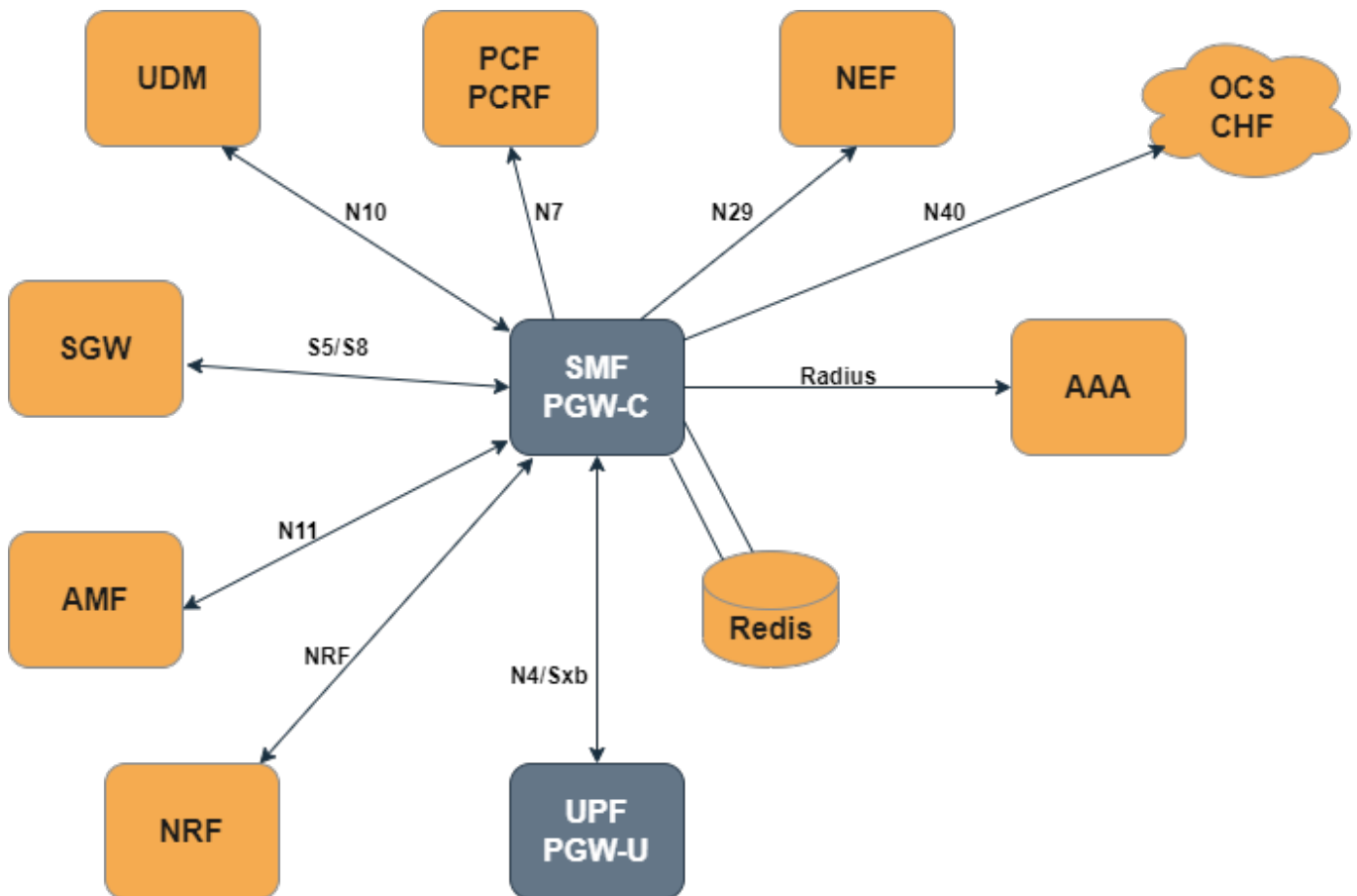


4. SMF Node

Session Management Function (SMF) node in 5G or Packet Gateway - control part (PGW-C) in 4G CUPS configuration manages the sessions in the user plane of the core network.

In 5G SMF is responsible for establishing, modifying, and releasing sessions (which includes allocating and managing the IP addresses used by the UEs and handling the session state throughout its lifecycle). It maintains session information, such as the UE IP address, tunnel info, and quality of service (QoS) parameters. For every session, SMF creates a separate context with a unique ID (one context per <session_id-imsi> pair). The SMF instructs through PFCP protocol the User Plane Function (UPF) on how to handle the user plane data, based on policies obtained from the Policy Control Function (PCF). The SMF interacts with the Authentication, Authorization, and Accounting node (AAA) to obtain external UE IP and perform GTP encapsulation.

Figure 11 - SMF Connectivity Diagram



4.1. Interfaces Description

Below, you can find brief descriptions of the SMF interfaces.

Reference point	Description
N4/Sxb	Interacts with UPF through PFCP protocol to enforce establishment, modification, and deletion of

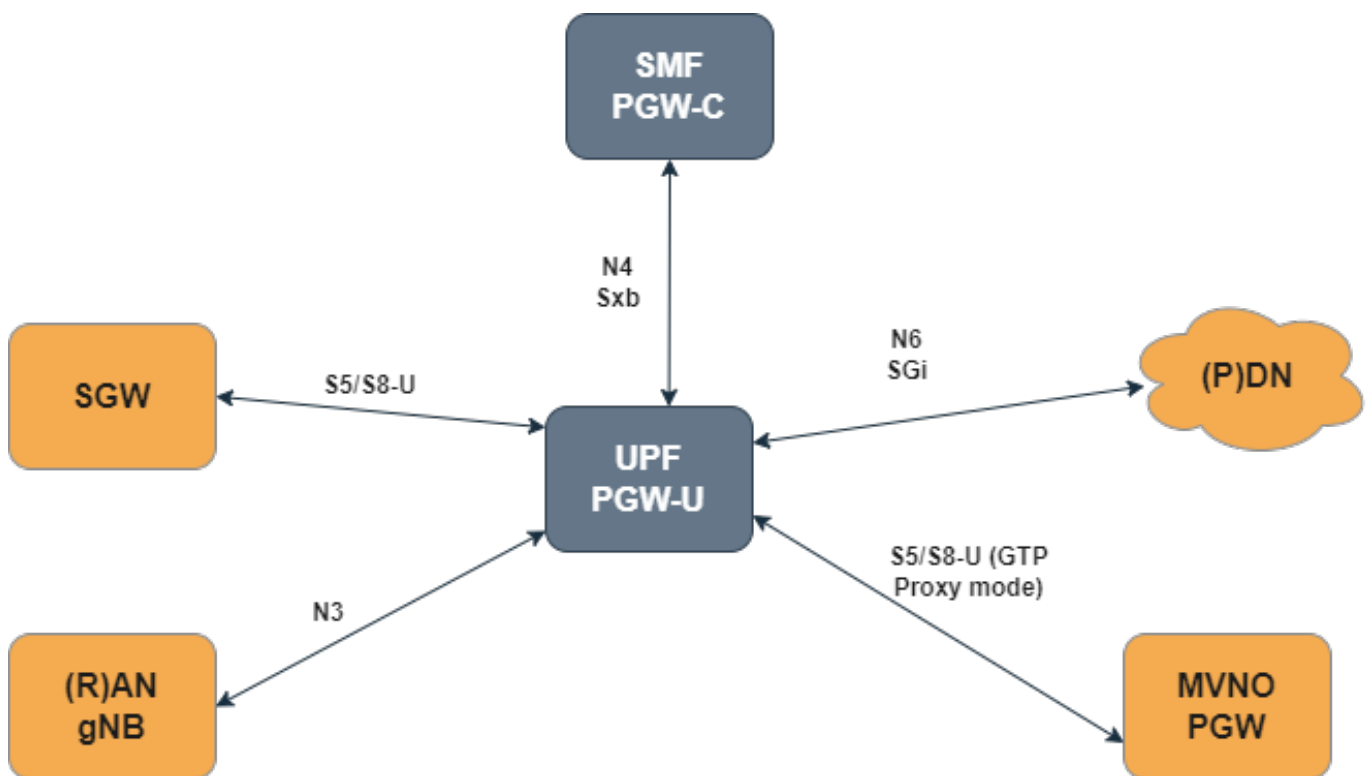
Reference point	Description
	session contexts. With PFCP, SMF sends packet detection and forwarding rules, as long as QoS rules, to the UPF.
N7	A service-based interface for communication with PCF, using HTTP2. The data is formatted as JSON. This interface triggers session management policies. SMF translates policies received from the PCF into a set of information and then forwards it to the UPF.
N10	A service-based interface for communication with UDM using HTTP2. The data is formatted as JSON. The interface is used from SMF to retrieve the UE's individual subscription data from the UDM.
N11	This interface is used for communication with AMF using HTTP2. It is used to forward N1 and N2 information from AMF to SMF for the creation, modification, and release of PDU Sessions.
NRF	The interface to the NRF is used from SMF to register in the NRF and to get the profiles of other registered nodes by the given criteria.
N29	Interface with NEF. Play a role in enabling NIDD services.
N40	This interface is used for communication from SMF to OCS/CHF.
Radius	A protocol is used for authentication, authorization, and accounting (AAA) in network access.
S5	Provides user plane tunneling and tunnel management between the SGW and the PDN-GW. It is used for SGW relocation due to UE mobility and if the SGW needs to connect to a non-collocated PDN-GW for the required PDN connectivity.
Gx	Provides QoS policy and charging rules transfer from the PCRF to the PCEF in the PDN-GW
S8	Inter-PLMN reference point providing user and control plane between the SGW in the VPLMN and the PDN-GW in the HPLMN. S8 is the inter-PLMN variant of S5.
SGi	The reference point between the PDN-GW and the PDN, which may be an operator's external public or private PDN or an intra-operator PDN, e.g., for provisioning of IMS services. The reference point corresponds to Gi for 3GPP accesses.
Gy	Online charging interface is used in mobile core networks. It connects the PGW/SMF (in LTE/5G) to the Online Charging System (OCS).

5. UPF Node

The UPF node in 5G (PGW-U in EPC) processes user traffic. It performs the following actions:

- Establishes, modifies, and deletes user traffic sessions when it receives commands from the SMF instructing it to do so
- Encapsulates / Decapsulates UE traffic in/from GTP-U packets
- Applies QoS parameters
- Reports the volume of used traffic during a session to the SMF
- Enforcement of PCC rules like throttling and allowlist

Figure 12 - UPF Connectivity Diagram



5.1. UPF Interfaces Description

The Packet Forwarding Control Protocol (PFCP) protocol is utilized on the following reference points (3GPP interfaces) described below:

Reference Point	Description
N4	This interface facilitates communication between the UPF and SMF in 5G networks.
Sxb	This interface enables communication between the PGW-C and PGW-U components in a CUPS-based EPC architecture.
N3	This access interface connects the UPF to the RAN node (usually a gNB) in 5G networks and

Reference Point	Description
	supports the QCI GTP-U extension for quality-of-service signaling.
S5/S8-U	This access interface links the SGW to the PGW-U in a CUPS-enabled EPC architecture, facilitating the transfer of user data.
S5/S8-U	This interface is used in GTP proxy mode to connect the PGW-U, functioning as a GTP-U proxy, with the MVNO's PGW. It is classified as a core network interface.
N6-AAA	This core network interface is used to connect the UPF to the AAA server in deployments that support PDN redundancy and static IP address assignment.
SGi-AAA	This core interface connects the PGW-U to the AAA server in scenarios involving PDN connectivity redundancy with static IP address allocation.
N6	This core interface in the 5G Core (5GC) architecture links the UPF to the external Data Network (DN), enabling user data transfer to internet or enterprise services.
SGi	In the EPC architecture, this interface connects the PGW-U to the Packet Data Network (PDN), serving as a core network interface.

5.2. UPF Features

Core-side encapsulation - GTP-U encapsulation on the Core interface (towards the DN). The SMF configures it wholly via PFCP.

Bitrate reset - periodically reset the MBR of sessions to a value that was previously modified using a PFCP QER.

- **TTL on SGi** - A user could configure their UE to use IP addresses from a private IP range, which would then be translated (using MASQUERADE) by the UE to its own UE IP. This allows the devices to access the PDN via the mobile network.

NOTE: To prevent two-way communication between other devices and the PDN, UPF could set the TTL of incoming downlink packets on SGi to zero.

6. Features

6.1. UPF Selection

SMF maintains a map indicating which UPF is responsible for each MCC or APN(s).

UPF sends the relevant selection information to the SMF in the PFCP Association Setup Request (ASR) message:

- If UPF is configured to be associated with some MCC and the Node ID IE is of type FQDN and contains MCC.
- If UPF is configured to be associated with the APN, it is defined in a separate IE.
- If UPF is not configured with any of the above parameters, it is used as a fallback for any session.

When the SMF receives a GTP-C Create Session Request, it selects a UPF based on the stored information for available UPFs, using the following order:

1. Search UPF for the requested APN/vAPN.
2. Search UPF for the MCC of the visited PLMN.
3. Select from the list of fallback UPFs.

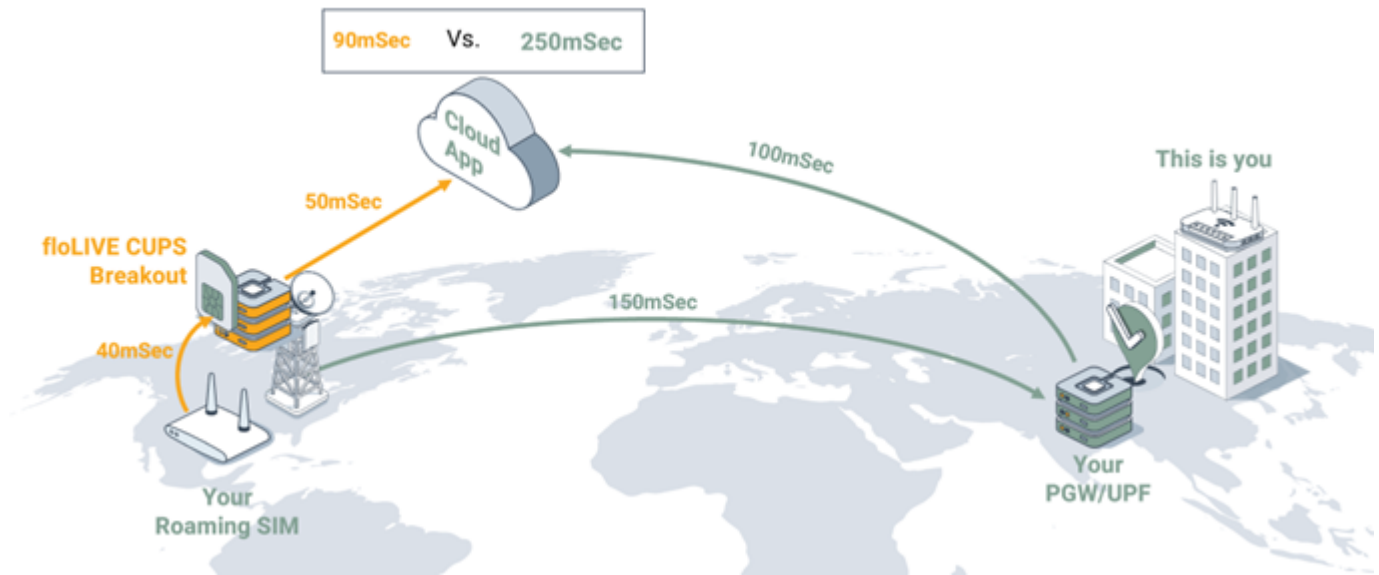
6.2. UPF Selection for Local Breakout Solution

The local breakout is the point of interconnection to external IP networks. This includes functionality for IP address allocation, charging, packet filtering, and policy-based control of user specific user-specific IP flows. It also plays a key role in supporting QoS for end-user IP services.

The user data payload - the IP packets flowing to and from the mobile devices - are handled by two logical nodes called the Serving Gateway (SGW) and the Packet Data Network Gateway (PDN-GW), where PDN is the Packet Data Network.

When deploying the local breakout on a remote location, then roaming traffic will not need to be directed to the originating MNO but remains in the current territory (the yellow line in the diagram below):

Figure 13 - Local Breakout Solution



- Providing local breakouts at the intersection points allows better control and visibility of the communication. This provides the following values: Performance – With roaming MNOs connecting to the local breakout, there is no need for the traffic to go through the home MNO, hence reducing the need for additional traffic and latency reduction.
- Localized Regulation – Complying with local regulation and ensuring the data does not traverse outside of the required region, to support GDPR and other relevant compliance standards.
- Localization – Using the local breakout component, the device can stay with the local region IP and not the home MNO one.
- Security – Reducing the transport and increasing control on the relevant hops to reduce risk for security vulnerabilities.

6.3. Lawful Interception

The Lawful Interception Gateway (LIGW) is a Network Element (NE) that combines the Lawful Interception Provisioning Function (LIPF) and MDF2/MDF3. It is responsible for NE discovery, task provisioning, and forwarding all the IRI to the real MDF2 of the MNO.

The LI_X2 interface between our network elements and the LIGW is implemented using Apache Kafka. The network elements produce IRIs in Kafka, which the LIGW then reads and sends using TLS to any relevant MDF2.

6.4. Online Charging (Gy Interface)

Core Network (CN) nodes can support standard Gy interface and Gy over HTTP for online charging.

If a connection to the Billing system fails, CN nodes store CDRs in a file and send them to the Billing system after the connection is restored.

6.5. GTP Proxy

NOTE: The GTP Proxy requires a license to operate.

This system supports GTP proxy functionality, allowing it to efficiently relay traffic between the Serving Gateway (SGW) and the Packet Gateway (PGW).

- When receiving traffic from the SGW, it behaves like a PGW to handle the session.
- When sending traffic to the PGW, it behaves like an SGW to maintain compatibility.

Additionally, it can dynamically select the correct PGW based on the Access Point Name (APN) requested by the user, ensuring traffic is routed to the right external network or service.

6.6. AAA Support

The PGW(C) supports AAA (Authentication, Authorization, and Accounting) functionality using the RADIUS protocol.

PGW(C) can send RADIUS Access Requests to the AAA server with user credentials and session attributes. A typical authorization request includes parameters like User-Name, User-Password, Service-Type, and network identifiers. These are formatted in a text file and submitted using the radclient tool, which is part of the freeradius-utils package on CentOS and similar distributions.

Additionally, PGW(C) listens for CoA or Disconnect requests on a configurable port. These requests, which may include the Acct-Session-Id, User-Name, and NAS-IP-Address, enable the AAA server to terminate or modify active sessions based on updated subscriber conditions. The parameters such as User-Name, shared secrets, and listening ports are fully configurable and will be detailed in the system configuration documentation.

6.6.1. APN Configuration

The APN may be configured by default or in dependence on other requirements:

Here are the supported cases on APN configurations:

- No username password – neither PGW nor EU has configured a password in the settings of the APN. Nothing will be sent.
- Always the same username and password – on PGW level, username and password are configured and will be static;
- The user behind the EU configures PAP and enters the username and password. PGW will overwrite and send those to Radius;
- The user behind the EU configures CHAP and enters the username and password. PGW will encrypt and send to Radius.

NOTE: PAP and CHAP authentication are performed at the device level. The customer should configure it on the device and enter the username and password.

- PGW supports the use of the User name as the MSISDN when needed.

6.6.2. Access Request

RADIUS Access Requests for authorizing session establishments such as Create Session or PDP Context requests, while also enabling the retrieval of subscriber IP addresses and enforcing billing policies

Supported AVP in Access Request

Authenticator: 372243bad45f21d38aae86c84cf6745a

AVP: t=User-Name(1) l=17 val=9725590000000000 //configuration in tbl_ggsn_apn_parameters can set the msisdn as username, e_msisdn_as_username = 10,

User-Name: 9725590000000000

NAS-IP-Address(4) l=6 val=0.0.0.0

Framed-Protocol(7) l=6 val=GPRS-PDP-Context(7)

Framed-IP-Address(8) l=6 val=10.128.0.1

Called-Station-Id(30) l=43 val=internet-billing-diameter-radius-acc-auth

Calling-Station-Id(31) l=17 val=9725590000000000

Acct-Session-Id(44) l=18 val=C0A860E73081BB01

NAS-Port-Type(61) l=6 val=Ethernet(15)

Vendor-Specific(26) l=23 vnd=3GPP(10415)

3GPP-IMSI(1) l=17 val=222010214146418

Mobile Country Code (MCC): Italy (222)

Mobile Network Code (MNC): Telecom Italia Mobile (TIM) (01)

Vendor-Specific(26) l=12 vnd=3GPP(10415)

3GPP-Charging-ID(2) l=6 val=813808385

Vendor-Specific(26) l=12 vnd=3GPP(10415)

3GPP-PDP-Type(3) l=6 val=IPv4(0)

Vendor-Specific(26) l=31 vnd=3GPP(10415)

VSA: t=3GPP-GPRS-Negotiated-QoS-profile(5) l=25 val=08-02FF1111111122222222

UMTS GTP QoS Profile

Version: 0x08

Hyphen separator: -

Allocation/Retention Priority: 0x0002

QCI: 255

Uplink APN Aggregate Maximum Bit Rate: 286331153

Downlink APN Aggregate Maximum Bit Rate: 572662306

Vendor-Specific(26) l=12 vnd=3GPP(10415)

VSA: t=3GPP-SGSN-Address(6) l=6 val=192.168.96.239

Vendor-Specific(26) l=12 vnd=3GPP(10415)

VSA: t=3GPP-GGSN-Address(7) l=6 val=192.168.96.231

Vendor-Specific(26) l=9 vnd=3GPP(10415)

3GPP-NSAPI(10) l=3 val=5

Vendor-Specific(26) l=9 vnd=3GPP(10415)

3GPP-Selection-Mode(12) l=3 val=MS or network provided APN, subscribed verified

Vendor-Specific(26) l=13 vnd=3GPP(10415)

VSA: t=3GPP-SGSN-MCC-MNC(18) l=7 val=22201

AVP: t=Vendor-Specific(26) l=23 vnd=3GPP(10415)

VSA: t=3GPP-IMEISV(20) l=17 val=1234501000000000

3GPP-IMEISV: 1234501000000000

AVP: t=Vendor-Specific(26) l=9 vnd=3GPP(10415)

VSA: t=3GPP-RAT-Type(21) l=3 val=EUTRAN(6)

AVP: t=Vendor-Specific(26) l=10 vnd=3GPP(10415)

VSA: t=3GPP-MS-TimeZone(23) l=4 val=Timezone: GMT +0 hours 0 minutes No adjustment

AVP: t=Vendor-Specific(26) l=9 vnd=3GPP(10415)

VSA: t=3GPP-Allocate-IP-Type(27) l=3 val=Do-Not-Allocate(0)

6.6.3. Change of Authorization

Dynamic policy updates are facilitated via RADIUS Disconnect or Change of Authorization (CoA) messages. These can be triggered, for example, when a subscriber's quota is exhausted or their subscription is revoked.

need to add example of MSG, integration team can share

6.6.4. Accounting

RADIUS Accounting messages are used for usage monitoring and sometimes if Gy feature is not utilized, also for online charging billing.

Follow the link to review the [Attribute Value Pairs](#).

6.7. Supporting UE Static IP Allocation for Public and Private IPs

- Via HSS - Recommended option, since it can remove the AAA as another node that needs to support the service
- Via AAA

Dedicated bearer for VoLTE/IMS.

SMF UPF can be deployed on Kubernetes on-prem and a Kubernetes over EKS AWS.

6.8. IPv6 Support with Dual Stack

SMF can support IPv6 dual stack sessions, allowing the UE to connect with IPv6 and IPv4 stacks. Address allocation will be done dynamically by SMF that can support IPv6 Router Advertisement for IPv6 /64 prefix allocation.

Enable the system to handle both IPv4 and IPv6 network traffic, ensuring compatibility and future-proofing network operations.

To enable dual-stack functionality for IPv6 alongside IPv4, the necessary business requirements have been fulfilled:

- Deployed over Kubernetes
- Deployed in an AWS environment
- Supports dynamic IP allocation (CMPv6) and IPv6 Stateless Auto-configuration
- Static IP allocation (Roadmap feature)
- Supports multiple Gi IPs that can be on the same or dedicated interfaces
- Supports Traffic classification for packets with IPv6 for Rating Group
- Supports populating the UE IPv6 in CDR (including Ga interface CDRs)

6.9. Domain-Based Traffic Classification

The PGW/UPF is configured with a domain name and a TCP/UDP port. It resolves the domain to a list of IP addresses, which are then used for SPI (Service Provider Interface) communication.

6.10. Multi Gi

Operators face challenges in ensuring speed, security, and scalability. The floLIVE CORE Connectivity Management Platform solves this with the PGW/UPF Multi Gi feature, providing a future-proof, flexible solution that optimizes data flow and prepares networks for growing demands while maintaining seamless, secure operations.

Key Features

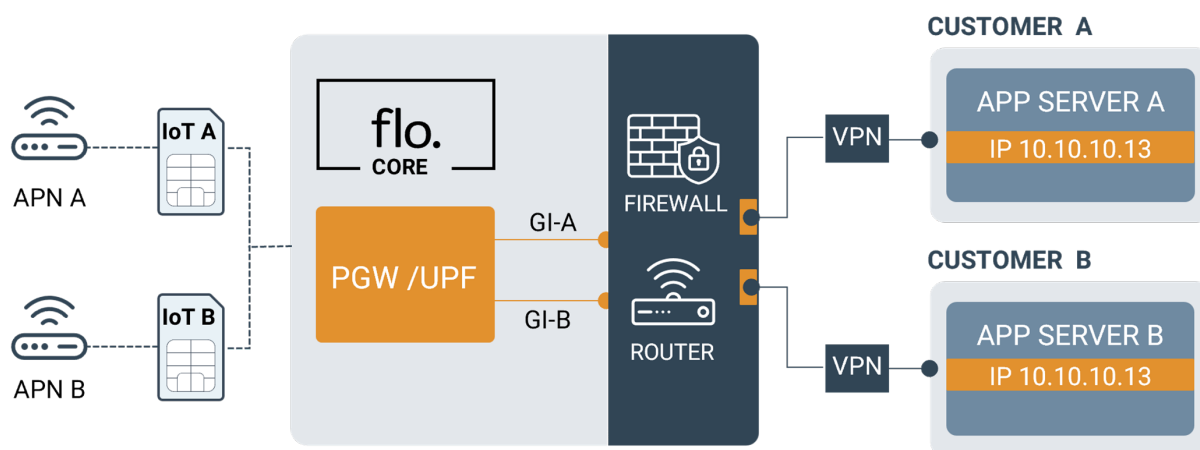
The key features of the floLIVE CORE PGW/UPF Multi Gi solution include:

1. **Multi-Gi Support:** Enables simultaneous connections for optimized bandwidth and traffic distribution, thereby reducing congestion and enhancing overall network performance.
2. **Seamless Integration:** It easily integrates with existing 2G, 3G, 4G, and 5G networks, causing minimal disruptions. This ensures a smooth upgrade path for network operators.
3. **Enhanced Security:** Robust encryption and security protocols protect data and ensure compliance with industry standards like GDPR and 3GPP.

How it Works

A block diagram of the floLIVE PGW/UPF Multi Gi solution is shown in this block diagram.

Figure 14 - Multi Gi Diagram



For PGW/UPF, multiple Gi interfaces can be defined, each configured to support several gateways. A combination of a Gi interface and a gateway is referred to as a "Network Instance." Each APN can be associated with a specific Network Instance, allowing for flexible configuration. For an APN with multiple Network Instances, Service Data Flow (SDF) filter rules can be configured to map APN and IP address filters to a Network Instance. Additionally, UPF can utilize various Network Instances based on APN and IMSI ranges through the UPF IMSI Analysis process.

6.11. TTL-Based Tethering Prevention

This feature allows the UPF to restrict the UE from providing connectivity to tethered devices through hotspot or alternative methods, by manipulating the TTL (Time-To-Live) value of IP packets on the SGi interface. It's designed to block two-way communication from hidden, non-UE devices masquerading behind the UE's IP address.

How It Works

Users may use hotspot option in the UE to route traffic from other devices through their own UE IP.

To mitigate this, the UPF can set the TTL of downlink IP packets to zero before forwarding them on SGi.

These packets still traverse the GTP path (SGW/eNB) as normal. Once it reaches the UE:

If destined for the actual UE, the packet is accepted and processed.

If routed to another device behind the UE, the zero TTL causes the packet to be dropped during forwarding.

This approach blocks tethered devices from receiving downlink data, breaking their session without impacting legitimate UE traffic.

Note: Although technically-savvy users may still manipulate TTL locally on the UE, this method raises the barrier significantly.

Configuration

Settings are managed in `tbl_configurations` under the PGWU/UPF section:

- `SGI_MODIFY_TTL`: Enables or disables TTL modification on SGi.
- `SGI_TTL`: Sets the TTL value applied to incoming packets (commonly set to 0 to enforce drop-on-forward).

Benefits

- Blocks unauthorized tethering behaviour at the UPF level.
- Requires no DPI or per-flow inspection.
- Transparent to legitimate users.
- Lightweight and scalable for high-throughput UPF environments.

6.12. DPDK

The UPF DPDK feature is engineered for ultra-low latency and high throughput, leveraging the Data Plane Development Kit (DPDK) for optimized packet processing via line card integration.

By incorporating DPDK, the UPF bypasses the traditional kernel network stack, enabling direct user-space packet handling that significantly enhances performance and scalability. This capability is particularly valuable in both 4G and 5G environments, where speed and efficiency are critical, allowing the system to support millions of simultaneous sessions with carrier-grade reliability.

With DPDK acceleration, our UPF delivers efficient CPU utilization, reduced jitter, and deterministic behavior even under heavy traffic loads.

6.13. NAT Integrated Functionality

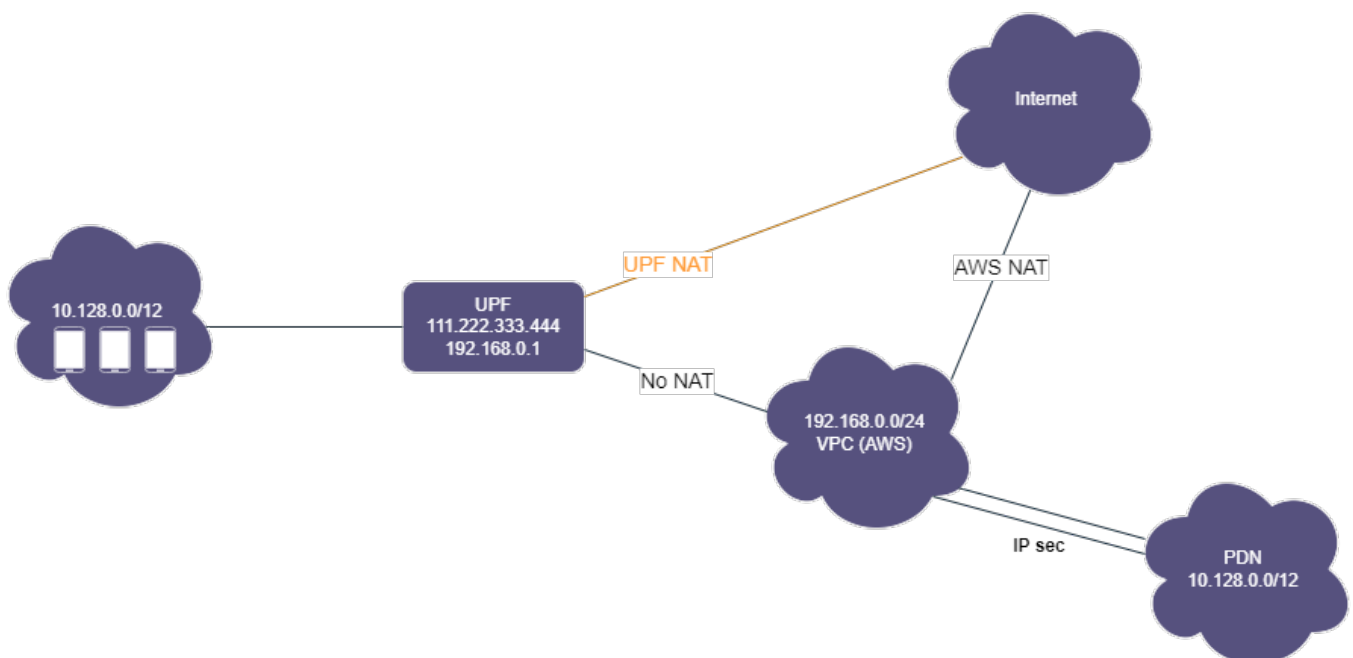
The purpose of this feature is to enable the UPF to forward UE traffic to the Internet and receive the traffic without the need for a separate NAT gateway for translation of UE IP addresses into routable addresses by the next hop (and vice versa).

NAT is performed only if the UE IPv4 address is private and the PDN host IP address is public (i.e., it is on the Internet).

The specific type of NAT that is implemented is IP tables MASQUERADE (or NAPT, as it is called in RFCs).

Stateful NAT of UE IPv4 is embedded in the UPF.

Figure 15 - NAT



NOTE: NAT is applied only on default SGi interface, not on additional interfaces in multi-Gi.

Private and Public Addresses

If an organization has a private IP network and a WAN link to a service provider. The private network's stub router is assigned a globally valid address on the WAN link. The remaining nodes in the organization have IP addresses that have only local significance. Nodes on the private network could be allowed simultaneous access to the external network, using the single registered IP address with the aid of NAPT. NAPT would allow mapping of tuples of the type (local IP addresses, local TU port number) to tuples of the type (registered IP address, assigned TU port number).

NOTE: NAT supports only one public IP address, allowing access for up to 64,000 users.

6.13.1. NAT Binding

A NAT binding describes the way internal IP addresses and ports are mapped to external ones and vice versa. It consists of the following things:

- An external port number.
- An entry in the SNAT map.
- An entry in the DNAT map.
- TTL (time to live), which is initialized, depending on the protocol is 10, 30 or 5 seconds after the time when the binding is created.

After each arrival of a packet in the UPF with a matching map entry (respectively in the SNAT or DNAT map) which is related to the binding, its TTL is reset to 10, 30 or 5 seconds in dependence of the protocol after the time of arrival (i.e., the binding is refreshed). Once the TTL expires without any packets matching the binding, the binding is deleted and stops participating in NAT.

6.13.2. NATP Packets Translation

Packets are translated as follows:

- In the uplink direction (i.e., during SNAT):
 - The source IP address and source port of TCP packets are translated to an external IP address and port.
 - The source IP address and source port of UDP packets are translated to an external IP address and port.
 - The source IP address and ICMP identifier of ICMP packets are translated to an external IP address and ICMP identifier.
- In the downlink direction (i.e., during DNAT):
 - The destination IP address and destination port of TCP packets are translated to an internal IP address and port.

- The destination IP address and destination port of UDP packets are translated to an internal IP address and port.
- The destination IP address and ICMP identifier of ICMP packets are translated to an internal IP address and ICMP identifier.

The transport layer identifiers, which include ports and ICMP identifiers, will be collectively referred to as ports.

NOTE: Other transport-layer protocols, besides TCP, UDP, and ICMP, are not supported.

6.14. APN Rate Control PCO

If GTP-C Create Session Request contains APN rate control element in the PCO or ePCO, SMF is sending APN rate control parameters in the response. Parameters are configured as string:

EXAMPLE: <time unit>-<rate value> , example: 3-100 , see 3GPP TS 24.008 section 10.5.6.3.2 APN rate control

Configuration options, starting with highest priority:

- Analysis action PCOARC (default string 3-100)
- APN parameter apn_rate_control
- Node configuration PGW - APNS_RATE_CONTROL

6.14.1. Define per APN or IMSI Range

In mobile core networks, the SMF (Session Management Function) may rely on external systems like CMP (Charging Management Platform) or a Rater to authorize or rate data sessions.

If the CMP/Rater does not respond, the SMF needs a predefined fallback behavior, which can be customized per APN or IMSI range:

- Allow
 - The SMF will allow the session to continue even if the CMP/Rater doesn't reply.
 - Used when service continuity is preferred over strict charging.
 - Often applied to critical services (e.g., emergency APNs or high-trust subscribers).
- Block
 - The SMF will block the session establishment or modification.
 - Used when charging info is mandatory before proceeding.
 - Common for prepaid users or services requiring strict control.

This configuration gives the operator fine-grained control over how to handle charging platform failures, balancing user experience and revenue protection.

6.15. Ga Interface

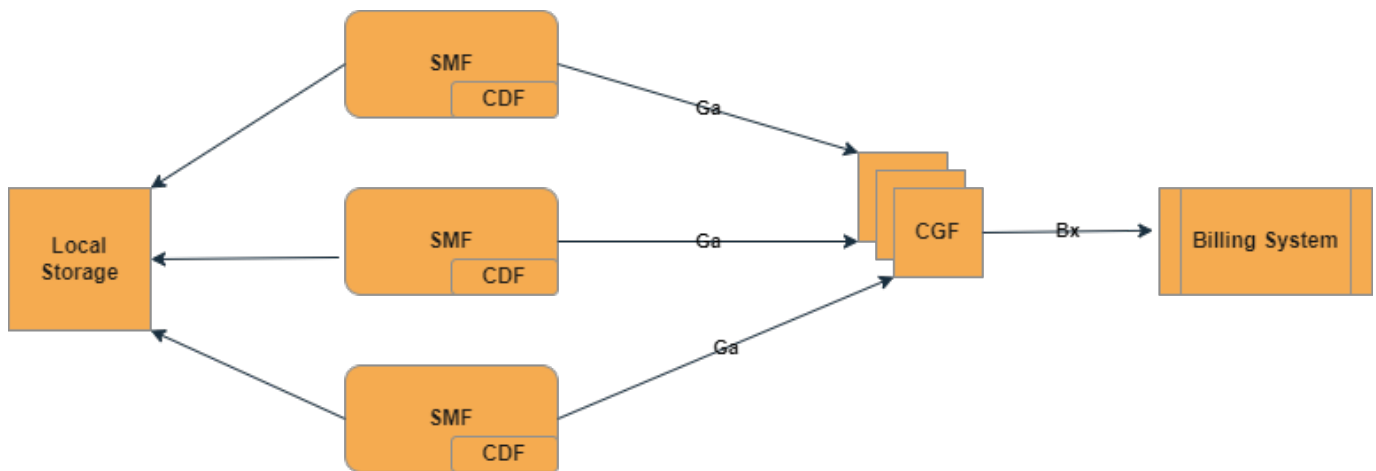
Ga is the interface for transferring CDRs between the individual elements of mobile core network that generate CDRs and the charging gateway, which further delivers the CDRs to billing system.

Charging Data Function (CDF) - function, performed at core network node, that is producing CDRs for reporting the usage;

Charging Gateway Function (CGF) - centralized point to collect the data and deliver it to billing system.

The message protocol on Ga interface is GTP' (gtp-prime), specified in 3GPP TS 32.298.

Figure 16 - Connectivity Diagram



The feature enables PGW to support the Rating Group on both the online and offline interfaces.

Rating group marking is set for different customer business categories, such as application, telemetric, and multimedia data streams.

6.15.1. CDR Generation in PGW via Ga Interface

The SMF is responsible for tracking and reporting user data session activity through Call Detail Records (CDRs). These records are generated and transmitted over the Ga interface toward the Charging Gateway Function (CGF), supporting both offline charging and usage-based billing.

CDR generation in the SMF is event-driven. The following triggers define when and how CDRs are created:

- Time-based
- Volume-based
- Session termination

6.15.1.1. Time-Based Trigger

Configuration

- Initiated when a new session is established.
- The interval for triggering CDR can be configured, such as setting it in seconds.
- The timer begins counting from session creation time.

Behavior upon each timer expiry:

- A CDR is generated.
- It includes the usage per rating group across all RGs since the last reported CDR.

Purpose

- Ensures periodic reporting for long-lived sessions.
- Supports data recovery in case of session interruption.
- Maintains billing accuracy by reducing data granularity loss.

6.15.1.2. Volume-Based Trigger

Configuration

- Defined per APN.
- Each rating group is monitored independently.

Behavior when a rating group crosses its volume threshold:

- A partial CDR is immediately created.
- The CDR includes usage only for the specific rating group that triggered it.
- Other groups are unaffected unless their individual thresholds are reached.

Purpose

- Provides fine-grained usage tracking for high-throughput applications.
- Enables granular charging policies by service or application type.

6.15.1.3. End-of-Session Trigger

Conditions

- Triggered when the session ends due to:
 - Explicit session release

- Timeout
- SMF or UE disconnection

Behavior of a final CDR that is generated upon session termination. It includes:

- All unreported usage.
- Across all rating groups with non-zero usage.

Purpose

- Ensures complete accounting of all session data.
- Finalizes usage for accurate offline charging.

Table: CDR Trigger Behavior

Trigger Type	Initiated By	CDR Includes	Scope
Time-based	Every X minutes	All rating groups (delta usage)	Partial
Volume-based	Threshold per rating group	Triggered rating group only	Partial
End of Session	Session teardown	All unreported usage (all groups)	Final

NOTE: Ga interface uses GTP' protocol for reliable record delivery.

NOTE: CDR sequence and integrity are ensured using record sequence numbering and intermediate buffering.

All CDRs are formatted according to 3GPP specifications (e.g., TS 32.298).

6.15.2. Volume-Based Partial CDRs on Ga Interface

This feature enables volume-based partial Call Detail Records (CDRs) on the Ga interface to enhance user data session activity tracking and reporting.

The SMF generates CDRs for user data sessions, which are transmitted over the Ga interface to the Charging Gateway Function (CGF). This supports offline charging and usage-based billing. CDR generation is event-driven, triggered by time, volume, or session termination.

The SMF is responsible for tracking and reporting user data session activity through Call Detail Records (CDRs). These records are generated and transmitted over the Ga interface toward the Charging Gateway Function (CGF), supporting both offline charging and usage-based billing.

6.15.3. Time-Based Partial CDRs on Ga Interface

The feature is implemented to support partial CDRs. CDR data includes only the most essential or available information at a specific point in time.

The limits are configured on the APN level.

6.15.4. Offline CDRs

Offline CDRs for SMF (PGW-C) are always written on session termination and after generating the usage report (every 24 hours).

Offline CDRs are in CSV format.

6.15.4.1. Ga Interface for Offline Charging

This feature enables the SMF/UPF to connect with its existing Jasper CMP for both online and offline charging.

Offline Call Detail Records over Ga:

- Offline CDR Transfers are done over TCP port 3386.
- Node Alive transactions are done over UDP port 3386.
- The CDR requires GTP Prime TLVs as defined by 3GPP TS 28.328 for PGW Records.

6.16. Policy Control and User Plane

6.16.1. Charging Interface

PGW supports the messages and functions required for charging subscribers.

The floLIVE PGW supports the following billing protocols: Diameter, Radius, and a PGW Billing HTTP API to support prepaid GPRS subscribers; it is a solution that enables the GGSN to constantly communicate with a prepaid billing server and dynamically request data volume quotas.

The operator's billing server is the node containing information about the subscriber's credit state. The API provides a mechanism to disconnect the subscriber's GPRS PDP session when the credit for data services has been exhausted.

The PGW implements Diameter according to 3GPP 32.299 (*Telecommunication management - Charging management - Diameter charging applications).

The API is based on the HTTP protocol, XML and works in a client-server mode. The following roles are defined:

- **Client (PGW):** The GGSN acts as a NAS, providing mobile data services to subscribers. The PGW dynamically requests volume quotas from the billing server. The PGW is responsible for tracking data usage, reporting it to the billing server, and tearing down the session when the billing instructs it.
- **Server (billing server):** The billing server contains all relevant subscriber information related to mobile data services (permissions, credit and other). It acts as a server that grants volume quotas upon the client's (GGSN) request. The billing server can return the unused quotas to a subscriber's account (e.g. if the subscriber closes the session before using the already granted quota).

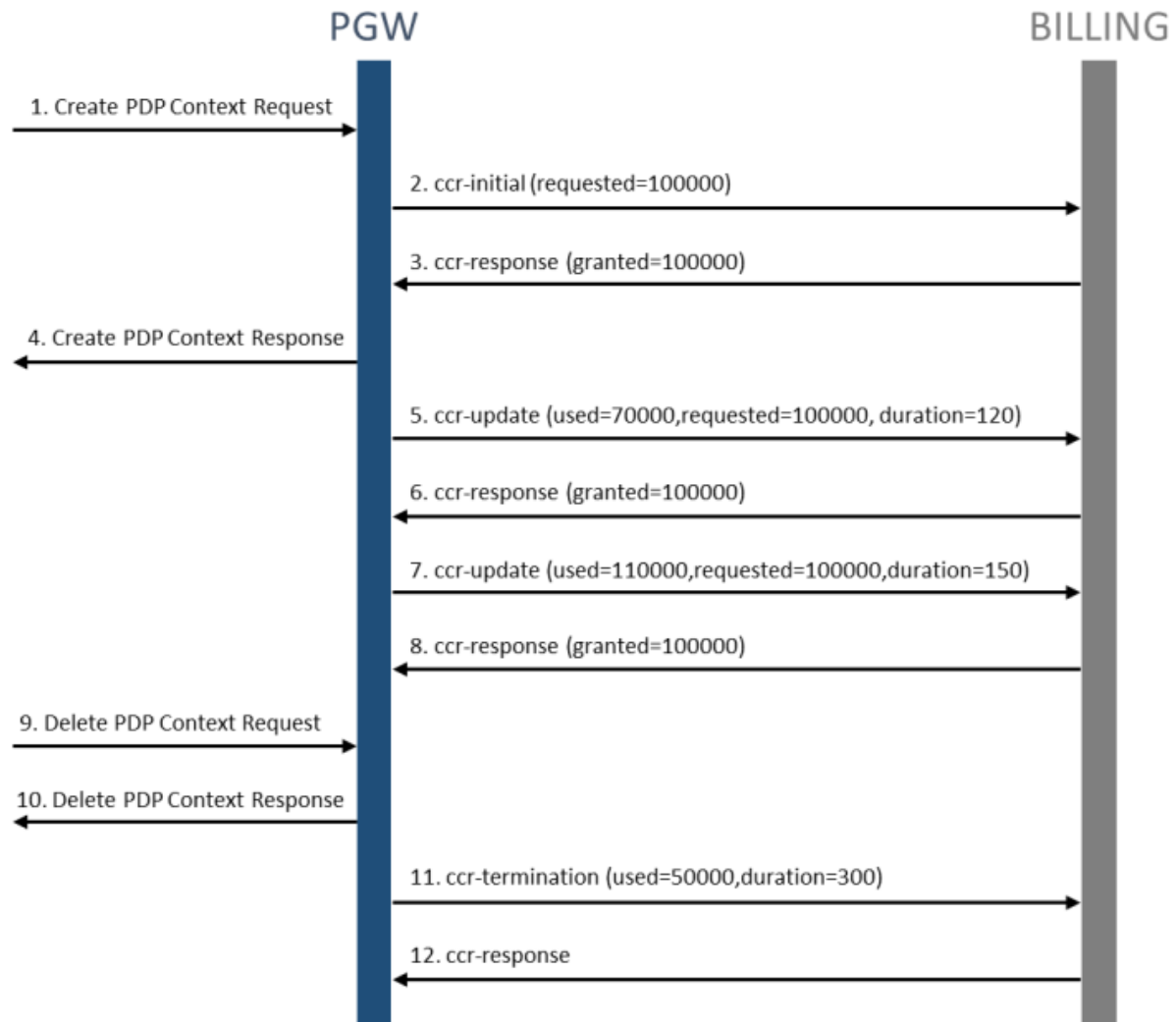
6.16.2. Billing Message Format

Billing messages use three types of request messages:

- **CCR Initial (ccr-initial):** Sent to the billing server at the start of a session. Used to initialize a session at the billing server and request the initial volume quota for the subscriber.
- **CCR Update (ccr-update):** Send to the billing server during the lifetime of the session when the current quota of the subscriber reaches a certain threshold (configured at the GGSN, e.g., 30% of the requested amount). Used to report data usage and to request additional volume quota.
- **CCR Termination (ccr-termination):** Sent to the billing server at the end of a session - if the subscriber disconnects manually, if the subscriber exhausts his/her credit for data services or if an error occurs. Used to report data usage and session termination.

The following diagram describes a flow of a session that is terminated by a subscriber:

Figure 17 - Billing Credit Control Flow

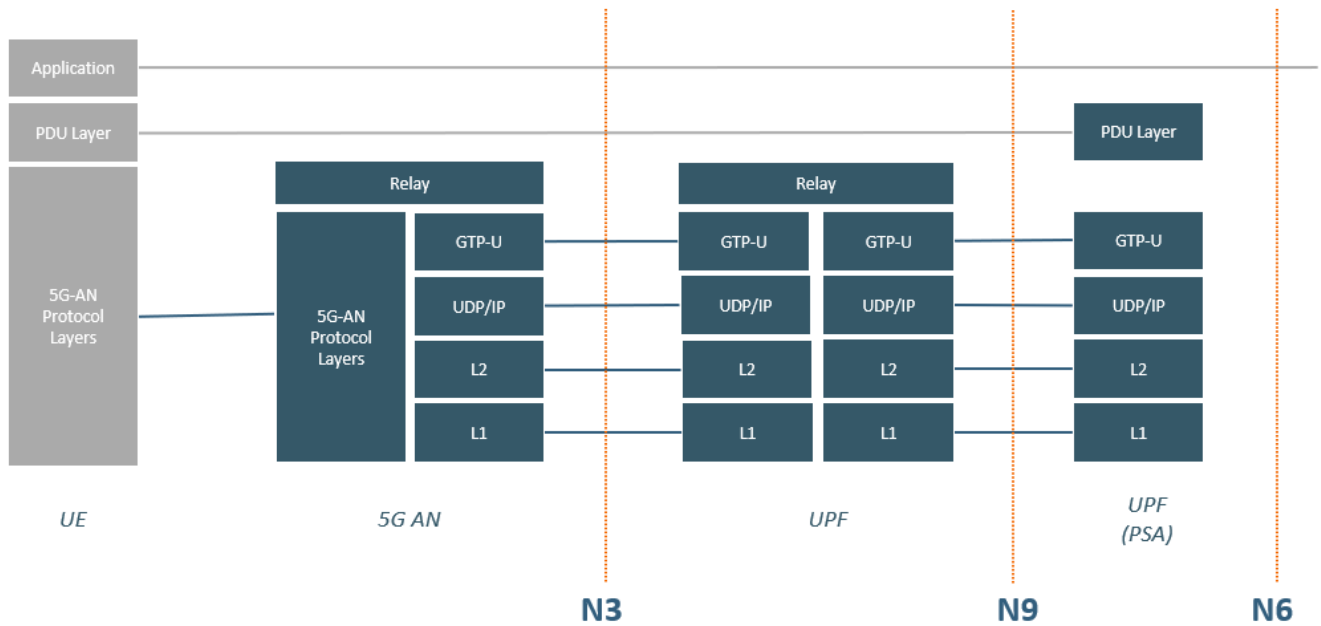


6.16.3. User Plane Handling

A direct session can be established from the AN (rand) and then a User Plane connecting towards UPF in the core network (over the **N3** reference point). Additional contact points are extended in the face of included UPFs (over the **N9** reference point) before the User Plane arrives at the Data Network (over the **N6** reference point).

Due to flexibility with GTP-U that existed in 4G, 5G UPF outperformed support to new requirements.

Figure 18 - User Plane Protocol Stack

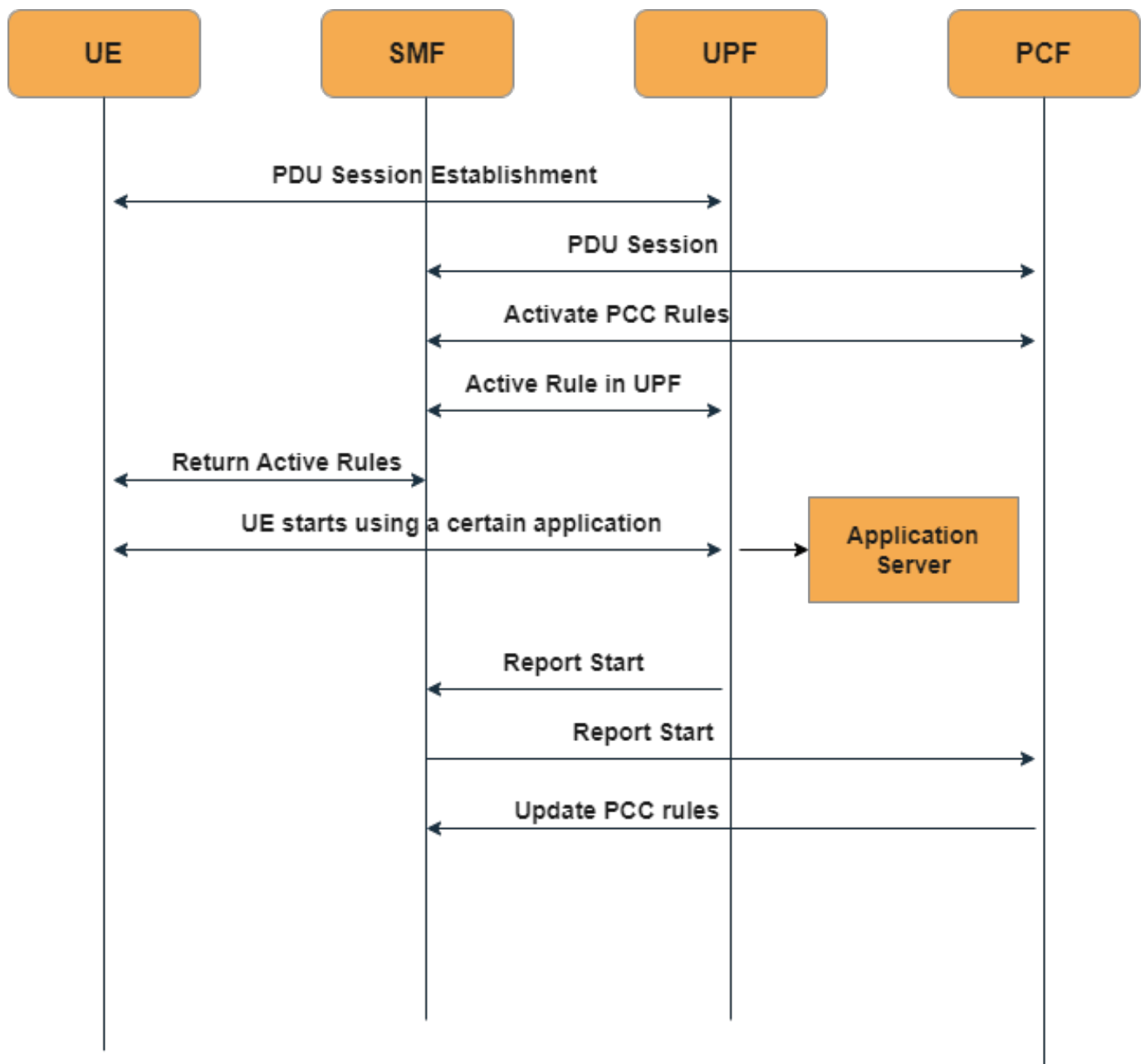


6.16.4. Traffic Steering Control

UPF performance on offload traffic is triggered via SMF to lead based on PCC rules:

- Network operator's policies
- Subscription action plan
- Selected current Radio Access Technology
- Network load status
- Application identifier
- Time of day
- Device location

Figure 19 - Activation on PCC Rule



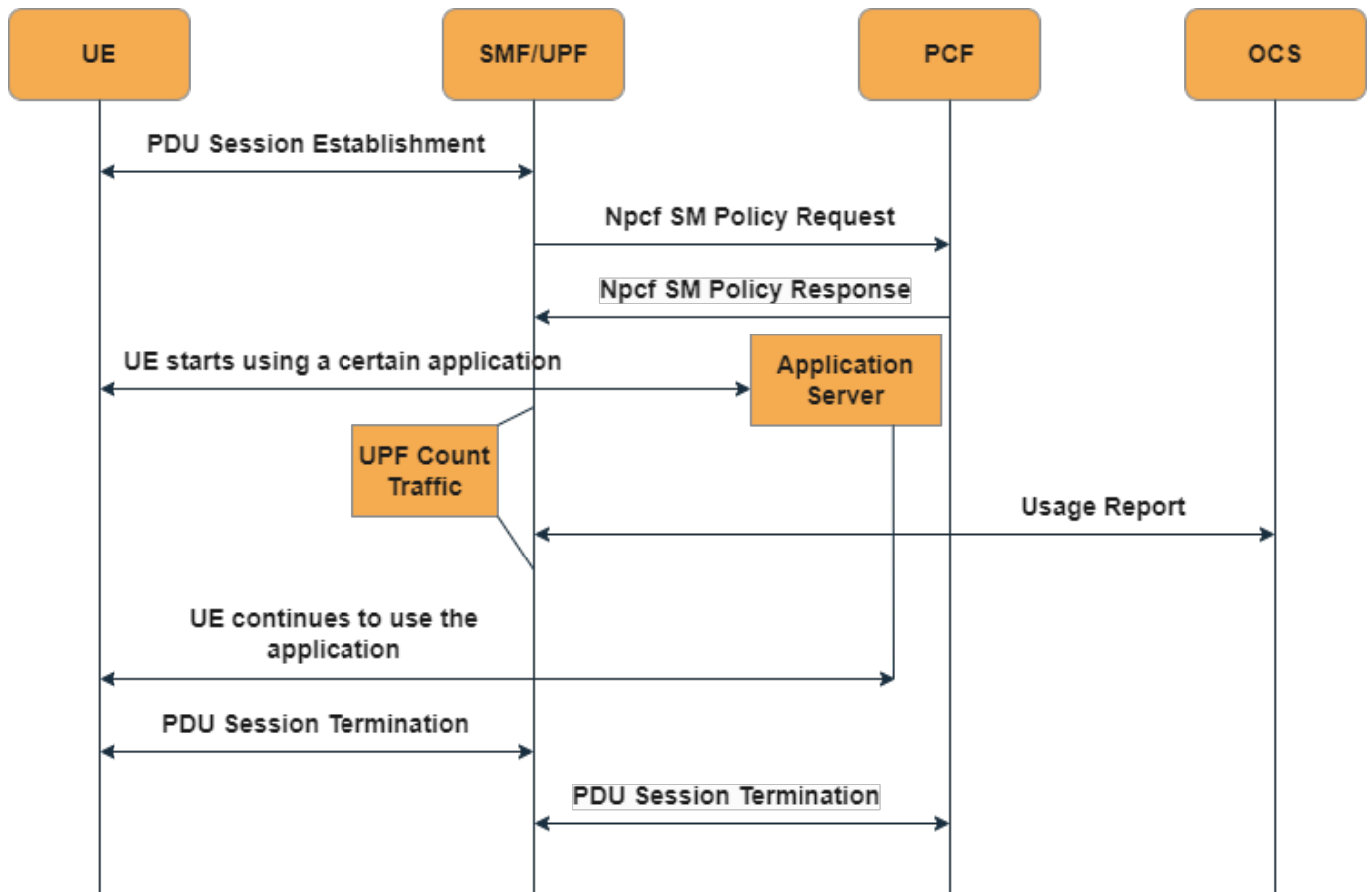
6.16.5. Usage Monitoring Control

Dynamic policy decisions based on the total network usage allow monitoring for volume and time thresholds on individual applications /services or groups of applications /services.

The accumulated usage is integrated into the UPF consideration for inspection on PDU sessions on which the traffic has direct reporting to SMF.

The message byte volume is sent from the SMF to the Online Charging System (OCS).

Figure 20 - Usage Monitoring Mapping of UPF



6.16.6. Policy Control

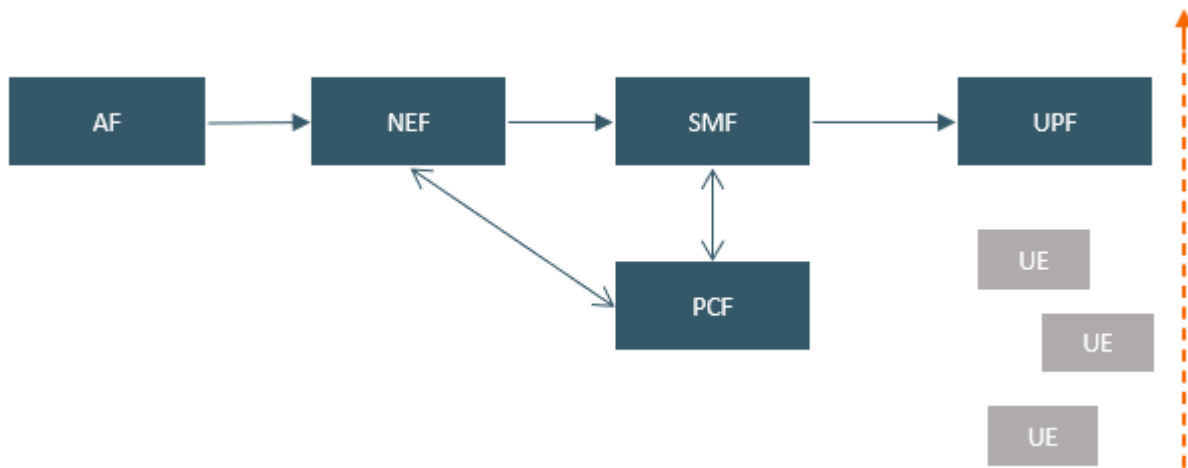
UPF opens an opportunity for classification using packet filtering that operates all IP data flows and facilitates Policy and Charging Control (PCC).

A part of the PCC functionality is the flow-based IP recognition, which includes support for service authorization and Quality of Service.

PCC—allows external applications to control various aspects of data sessions.

NOTE: You can import policy rules on which the UPF will act.

Figure 21 - Policy Control



7. PGW, SMF, and UPF Monitoring

The PGW, SMF, and UPF support integration with a centralized monitoring system that continuously scans all resources and notifies system and network administrators upon outages, malfunctions, disconnections and more.

Periodic as well as on-demand performance reports are generated to enable quick and efficient drill-down and route-cause analysis.

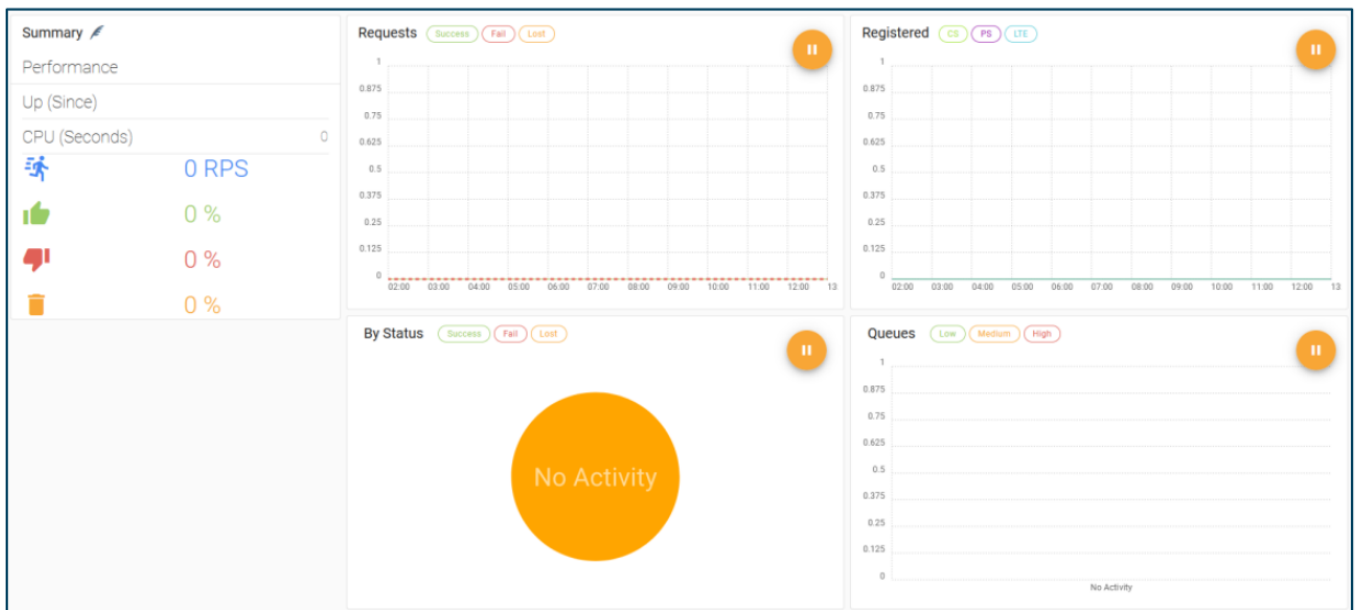
The PGW, SMF, and UPF support both SNMP and Traps methods:

- **Server and service resources:** CPU, memory, storage.
- **Service status:** Up / Down / Restart.
- **Statistics:** Open sessions, active resources, active/new bearer, PDP.

NOTE: Traps are sent both upon a system event and periodically (e.g., every 10 seconds).

The image below depicts a typical monitoring screen:

Figure 22 - Centralized Monitoring



7.1. PGW, SMF, and UPF Management Tools

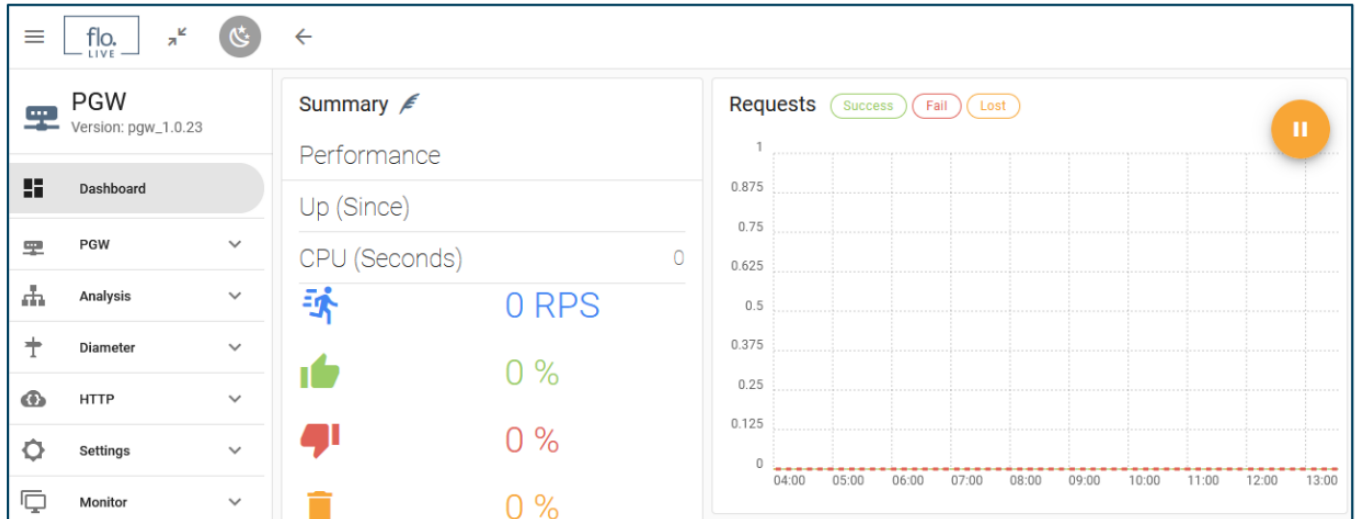
The PGW, SMF, and UPF supports a web-based graphical user interface tools to manage the following aspects:

- **IP Pools:** Managing local IP pools
- **Routs:** Managing the Radius flow routes
- **Routing:** Managing the different pools routing and redirections rules

- **APNs:** Managing system APNs settings and IP pools mapping
- **APN Routs:** Managing the connections between APNs and Radius routes

The image below depicts the PGW, SMF, and UPF Web Management tools:

Figure 23 - Web Management



8. Appendix 1

8.1. Attribute Value Pairs

[Click here to return to the Accounting page.](#)

AVP: t=Acct-Status-Type(40) l=6 val=Start(1)

AVP: t=NAS-IP-Address(4) l=6 val=0.0.0.0

Type: 4

Length: 6

NAS-IP-Address: 0.0.0.0 (0.0.0.0)

AVP: t=Acct-Delay-Time(41) l=6 val=0

AVP: t=Acct-Session-Id(44) l=18 val=C0A860E73081BB01

Type: 44

Length: 18

Acct-Session-Id: C0A860E73081BB01

AVP: t=Called-Station-Id(30) l=43 val=internet-billing-diameter-radius-acc-auth

Type: 30

Length: 43

Called-Station-Id: internet-billing-diameter-radius-acc-auth

AVP: t=User-Name(1) l=17 val=9725590000000000

Type: 1

Length: 17

User-Name: 9725590000000000

AVP: t=Service-Type(6) l=6 val=Framed(2)

Type: 6

Length: 6

Service-Type: Framed (2)

AVP: t=Framed-Protocol(7) l=6 val=GPRS-PDP-Context(7)

Type: 7

Length: 6

Framed-Protocol: GPRS-PDP-Context (7)

AVP: t=Framed-IP-Address(8) l=6 val=100.64.0.10

Type: 8

Length: 6

Framed-IP-Address: 100.64.0.10 (100.64.0.10)

AVP: t=Calling-Station-Id(31) l=17 val=9725590000000000

Type: 31

Length: 17

Calling-Station-Id: 9725590000000000

AVP: t=Acct-Authentic(45) l=6 val=RADIUS(1)

AVP: t=NAS-Port-Type(61) l=6 val=Ethernet(15)

Type: 61

Length: 6

NAS-Port-Type: Ethernet (15)

AVP: t=Vendor-Specific(26) l=23 vnd=3GPP(10415)

Type: 26

Length: 23

Vendor ID: 3GPP (10415)

VSA: t=3GPP-IMSI(1) l=17 val=222010214146418

Type: 1

Length: 17

IMSI: 222010214146418

[Association IMSI: 222010214146418]

Mobile Country Code (MCC): Italy (222)

Mobile Network Code (MNC): Telecom Italia Mobile (TIM) (01)

AVP: t=Vendor-Specific(26) l=12 vnd=3GPP(10415)

Type: 26

Length: 12

Vendor ID: 3GPP (10415)

VSA: t=3GPP-Charging-ID(2) l=6 val=813808385

Type: 2

Length: 6

3GPP-Charging-ID: 813808385

AVP: t=Vendor-Specific(26) l=12 vnd=3GPP(10415)

Type: 26

Length: 12

Vendor ID: 3GPP (10415)

VSA: t=3GPP-PDP-Type(3) l=6 val=IPv4(0)

Type: 3

Length: 6

3GPP-PDP-Type: IPv4 (0)

AVP: t=Vendor-Specific(26) l=12 vnd=3GPP(10415)

Type: 26

Length: 12

Vendor ID: 3GPP (10415)

VSA: t=3GPP-SGSN-Address(6) l=6 val=192.168.96.239

Type: 6

Length: 6

3GPP-SGSN-Address: 192.168.96.239 (192.168.96.239)

AVP: t=Vendor-Specific(26) l=12 vnd=3GPP(10415)

Type: 26

Length: 12

Vendor ID: 3GPP (10415)

VSA: t=3GPP-GGSN-Address(7) l=6 val=192.168.96.231

Type: 7

Length: 6

3GPP-GGSN-Address: 192.168.96.231 (192.168.96.231)

AVP: t=Vendor-Specific(26) l=31 vnd=3GPP(10415)

Type: 26

Length: 31

Vendor ID: 3GPP (10415)

VSA: t=3GPP-GPRS-Negotiated-QoS-profile(5) l=25 val=08-02FF11111112222222

Type: 5

Length: 25

UMTS GTP QoS Profile

Version: 0x08

Hyphen separator: -

Allocation/Retention Priority: 0x0002

....0.. = Pre-emption Capability (PCI): Enabled

....00 00.. = Priority Level: 0

....0 = Pre-emption Vulnerability (PVI): Enabled

QCI: 255

Uplink APN Aggregate Maximum Bit Rate: 286331153

Downlink APN Aggregate Maximum Bit Rate: 572662306

AVP: t=Vendor-Specific(26) l=13 vnd=3GPP(10415)

Type: 26

Length: 13

Vendor ID: 3GPP (10415)

VSA: t=3GPP-SGSN-MCC-MNC(18) l=7 val=22201

Type: 18

Length: 7

3GPP-SGSN-MCC-MNC: 22201

AVP: t=Vendor-Specific(26) l=9 vnd=3GPP(10415)

Type: 26

Length: 9

Vendor ID: 3GPP (10415)

VSA: t=3GPP-NSAPI(10) l=3 val=5

Type: 10

Length: 3

3GPP-NSAPI: 5

AVP: t=Vendor-Specific(26) l=9 vnd=3GPP(10415)

Type: 26

Length: 9

Vendor ID: 3GPP (10415)

VSA: t=3GPP-Selection-Mode(12) l=3 val=MS or network provided APN, subscribed verified

Type: 12

Length: 3

.... ..00 = Selection mode: MS or network provided APN, subscribed verified (0)

AVP: t=Vendor-Specific(26) l=23 vnd=3GPP(10415)

Type: 26

Length: 23

Vendor ID: 3GPP (10415)

VSA: t=3GPP-IMEISV(20) l=17 val=1234501000000000

Type: 20

Length: 17

3GPP-IMEISV: 1234501000000000

AVP: t=Vendor-Specific(26) l=9 vnd=3GPP(10415)

Type: 26

Length: 9

Vendor ID: 3GPP (10415)

VSA: t=3GPP-RAT-Type(21) l=3 val=EUTRAN(6)

Type: 21

Length: 3

3GPP-RAT-Type: EUTRAN (6)

AVP: t=Vendor-Specific(26) l=10 vnd=3GPP(10415)

Type: 26

Length: 10

Vendor ID: 3GPP (10415)

VSA: t=3GPP-MS-TimeZone(23) l=4 val=Timezone: GMT +0 hours 0 minutes No adjustment

Type: 23

Length: 4

Timezone: GMT +0 hours 0 minutes No adjustment

[Click here to return to the Accounting page.](#)

9. Abbreviations

AAA – Authentication, Authorization, and Accounting

AMF – Access and Mobility Management Function

DPDK - Data Plane Development Kit

NAT - Network Address Translation

NEF – Network Exposure Function

OCS – Online Charging System

PCF – Policy Control Function

PFCP – Packet Forwarding Control Protocol

SMF – Session Management Function

UDM – Unified Data Management

UE – User Equipment

UPF – User Plane Function

10. Specifications

TS 29.212 Gx interface

TS 29.244 Sxb/N4 interface

TS 29.274 GTPv2-C protocol (S5/S8-C)

TS 29.502 Session Management Services (SMF)